(Demystifying?) Catalytic Deracemization Reactions

and

Access to enantioenriched atropisomers through biocatalytic deracemization

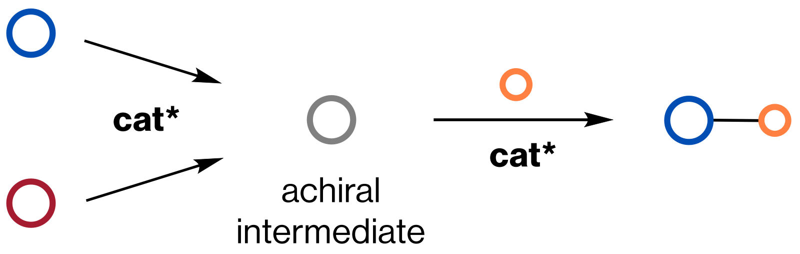
Casey B. Roos,[‡1] S. Luke Schulert, [‡1,2] Lara E. Zetzsche,[1,2] Spencer E. McMinn,[3] Angela E. Cheong,[1] Eunjae Shim,[1] Eugene E. Kwan,[3] and Alison R. H. Narayan[1,2,4*]

Literature Meeting

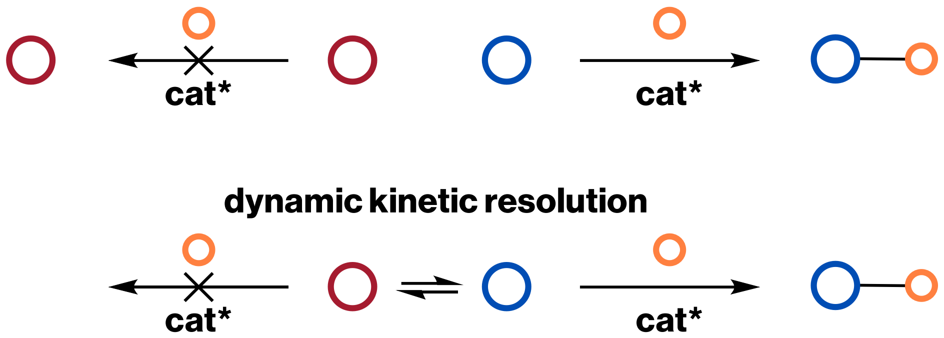
May 2025

Strategies for asymmetric catalysis from chiral starting materials are well-defined

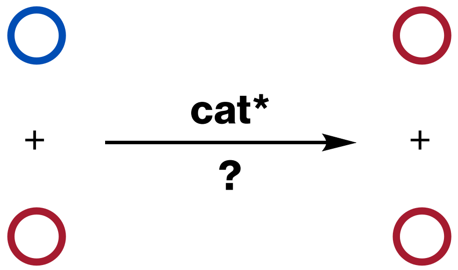
1 Proceed through prochiral intermediate/SM



2 Selective reaction from one enantiomer of SM



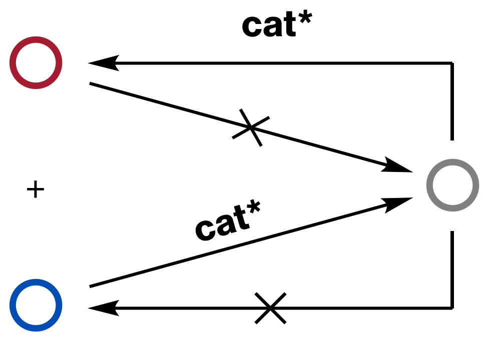
3 Racemate to single enantiomer



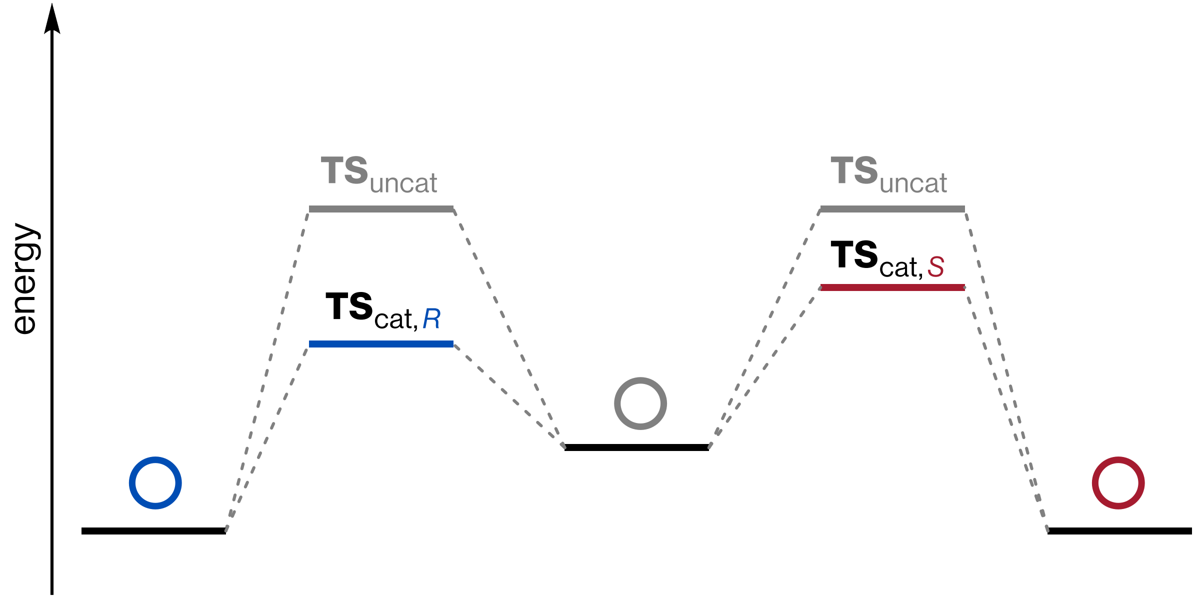
$$\Delta H = 0$$
$$\Delta G = 0 - T\Delta S$$
$$\Delta S = k_B \ln(2)$$
$$\Delta G(298K) = +0.42\ \text{kcal mol}^{-1}$$

Deracemization cannot be achieved in energetically closed systems

Closed system scheme:



The lowest path forward is also the lowest path back:



The rate-limiting barrier heights for (*S*) →(*R*) and (*R*) → (*S*) are identical.

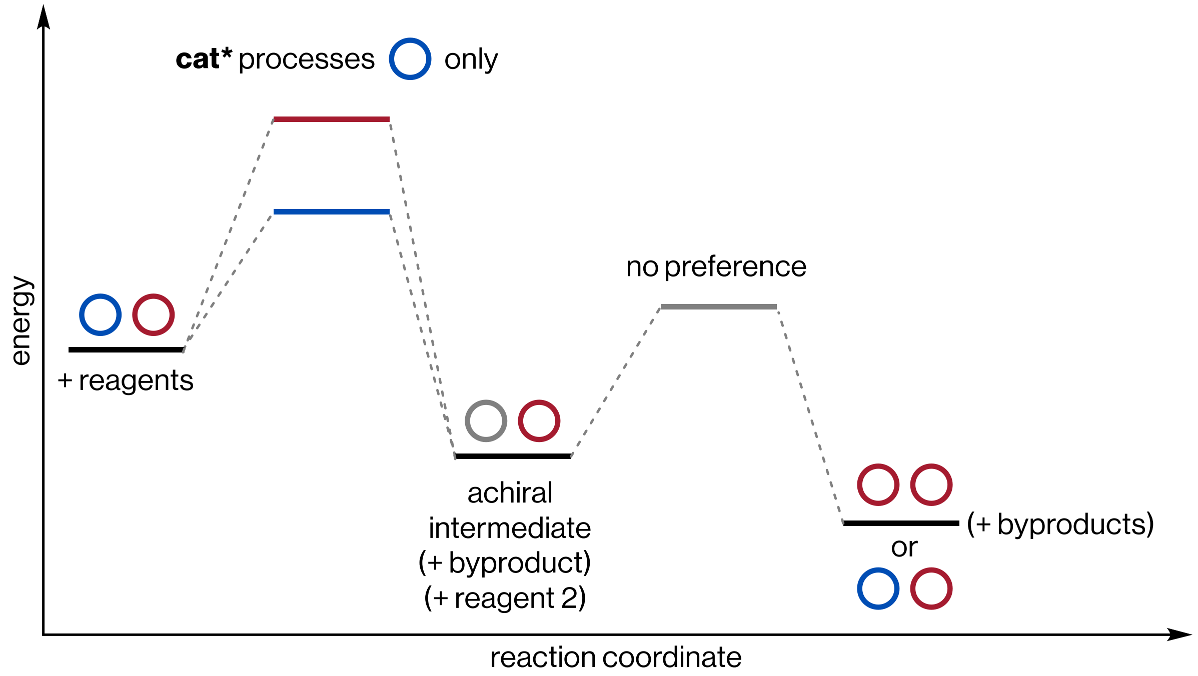
At equilibrium, we want a single chiral catalyst to selectively:

1. Deplete (*R*) ($k_{d,R} > k_{d,S}$)
2. Generate (*S*) ($k_{g,S} > k_{g,R}$)

In other words, the lower barrier to depleting (*R*) means that the intermediate also prefers to generate (*R*).

Forward/reverse reactions can be decoupled and rendered irreversible

Each step is a separate exergonic reaction



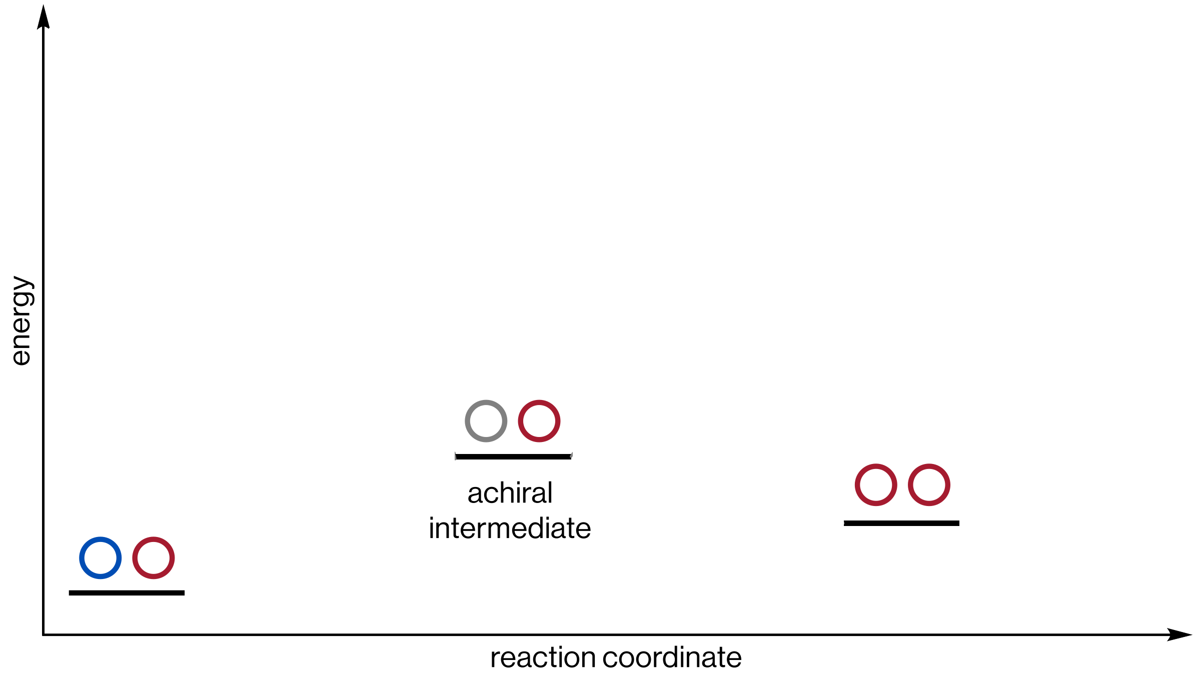
At the limit, this is a two-step reaction (isolate the intermediate)

Each cycle consumes two stoichiometric chemical reagent
e.g. oxidant and reductant

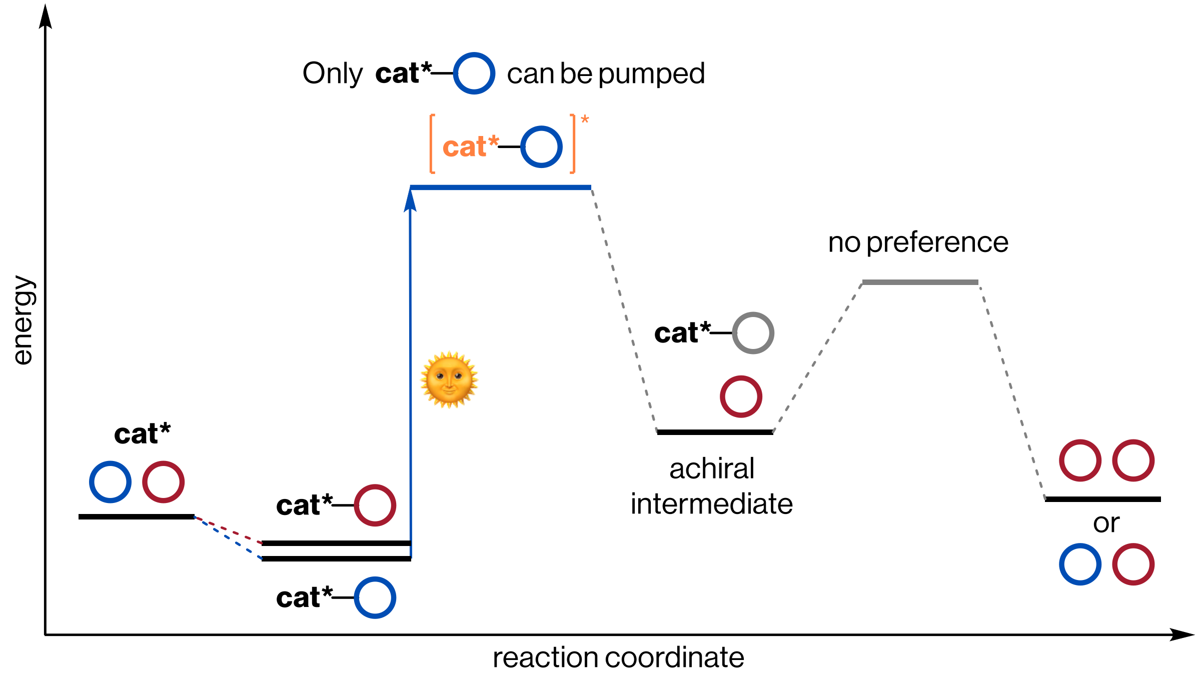
Annihilation of reagents is a common failure point

Excited-state chemistry offers adjacent strategies

Pump the starting materials



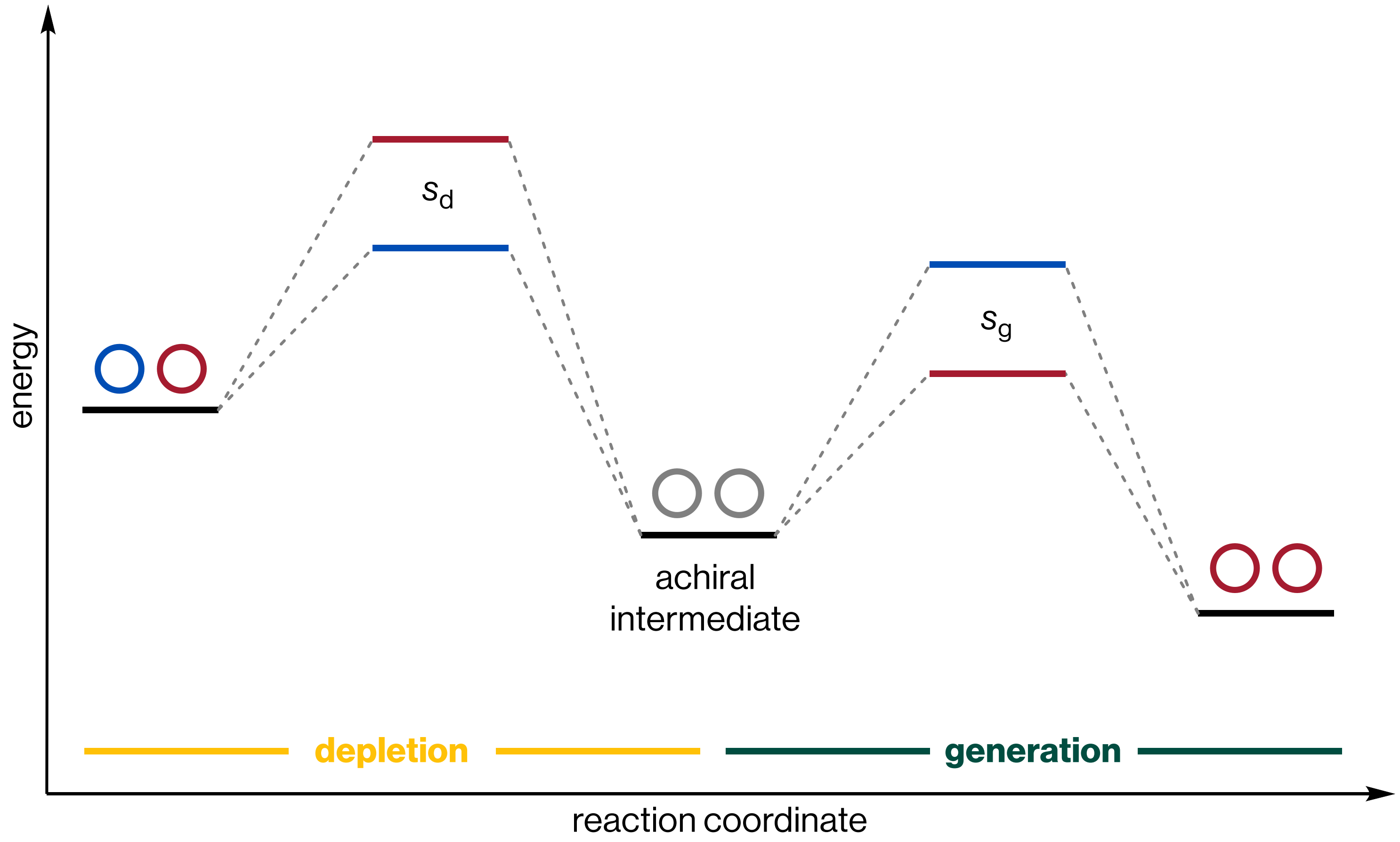
Pump the catalyst-SM complex



Excitation of either the chiral catalyst, the substrate, or the catalyst–substrate complex is possible.

Each cycle consumes stoichiometric energy (photon)

Selectivity in deracemization can be considered through depletion/generation

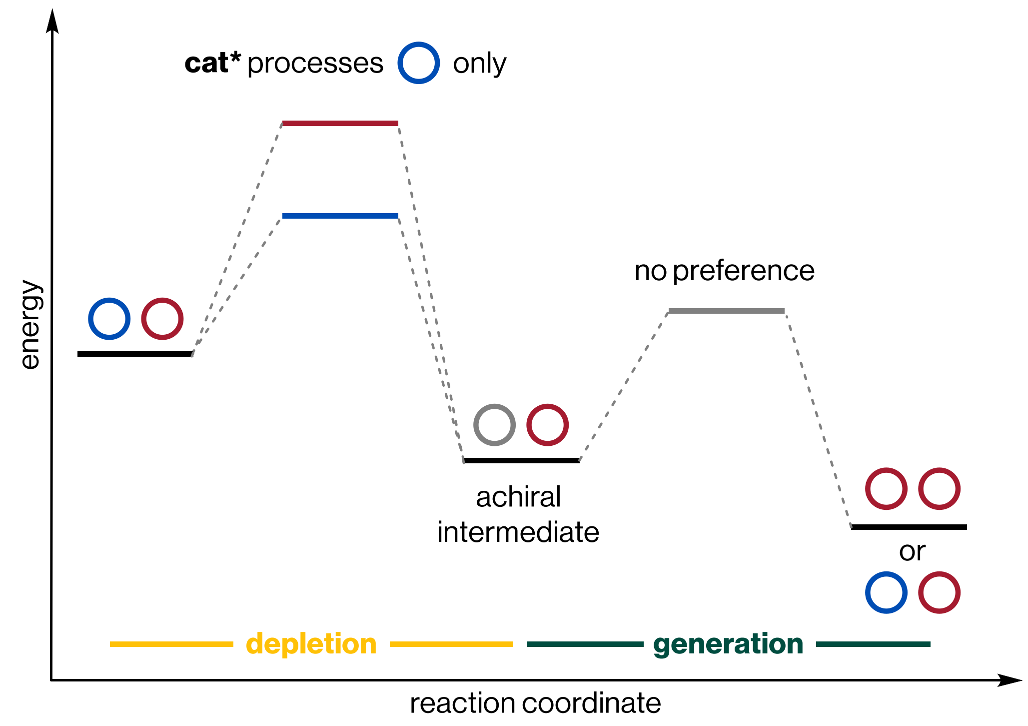


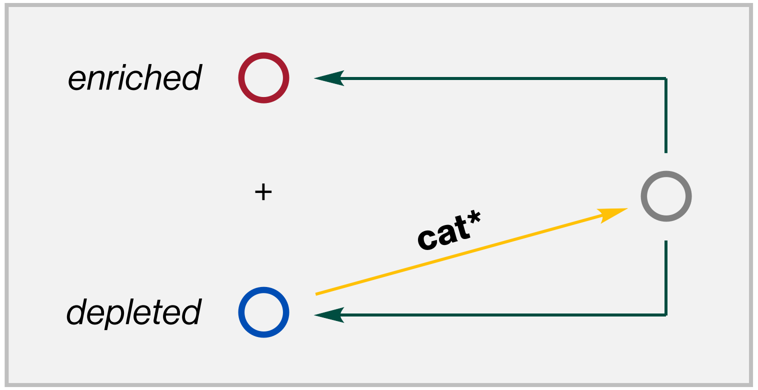
Selectivity can enforced in only the **depletion** phase, in only the **generation** phase, or in both.

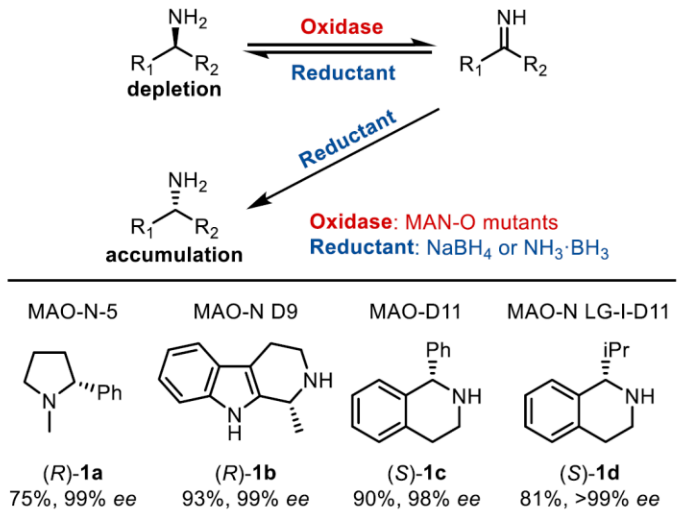
Luo *JACS* **2023**, *145*, 10917

Cyclic deracemization: selective depletion

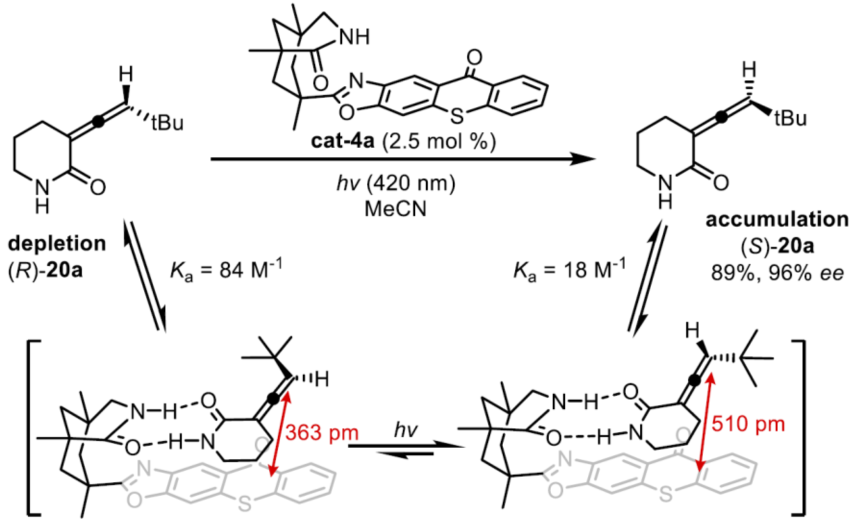
1 Cyclic deracemization







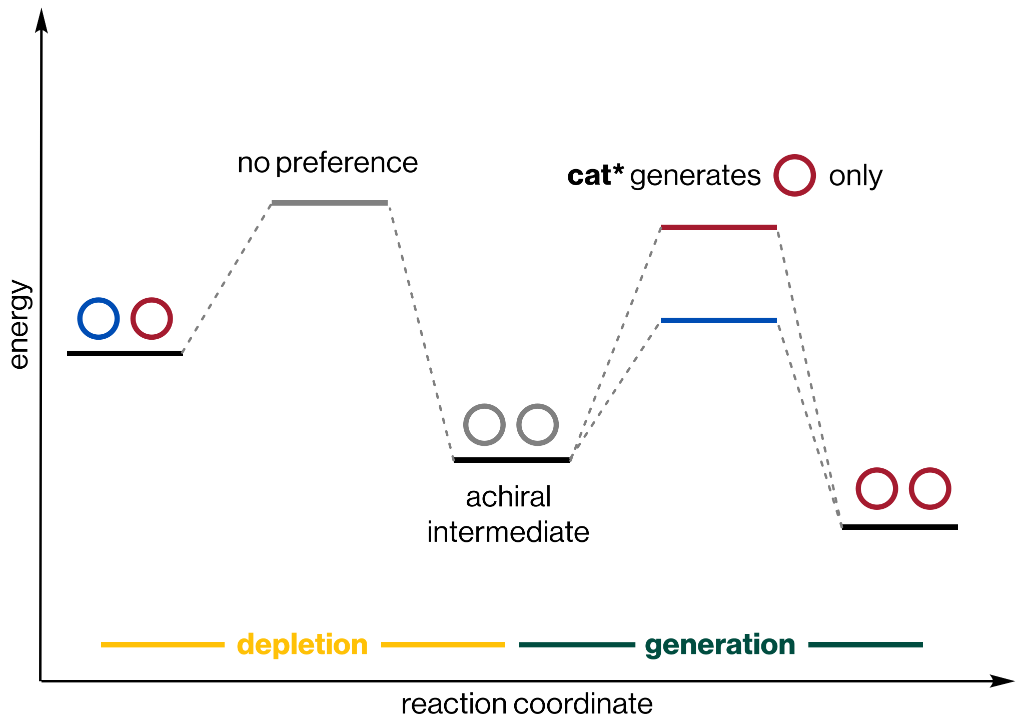
Turner *ACIE* **2002**, *41*, 3177

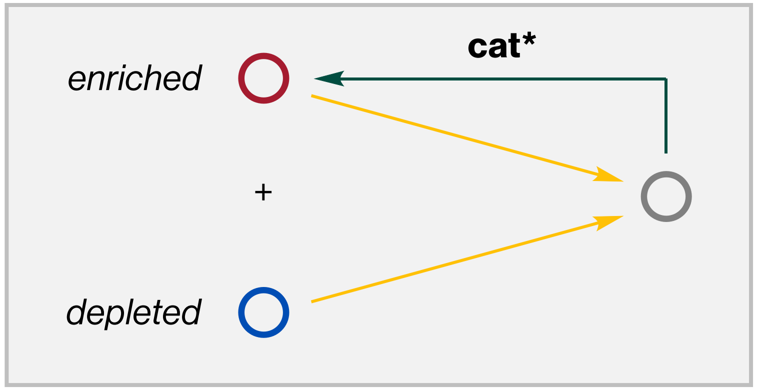


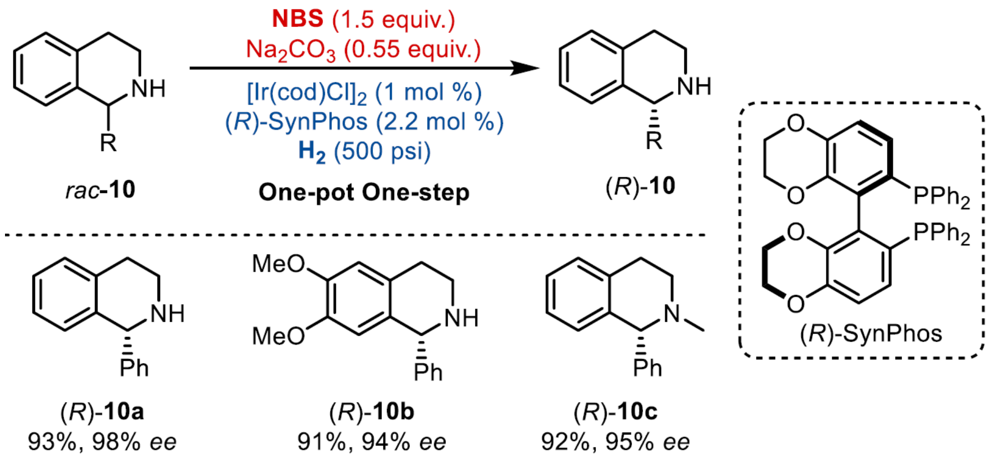
Bach *Nature* **2018**, *564*, 240

Linear deracemization: selective generation

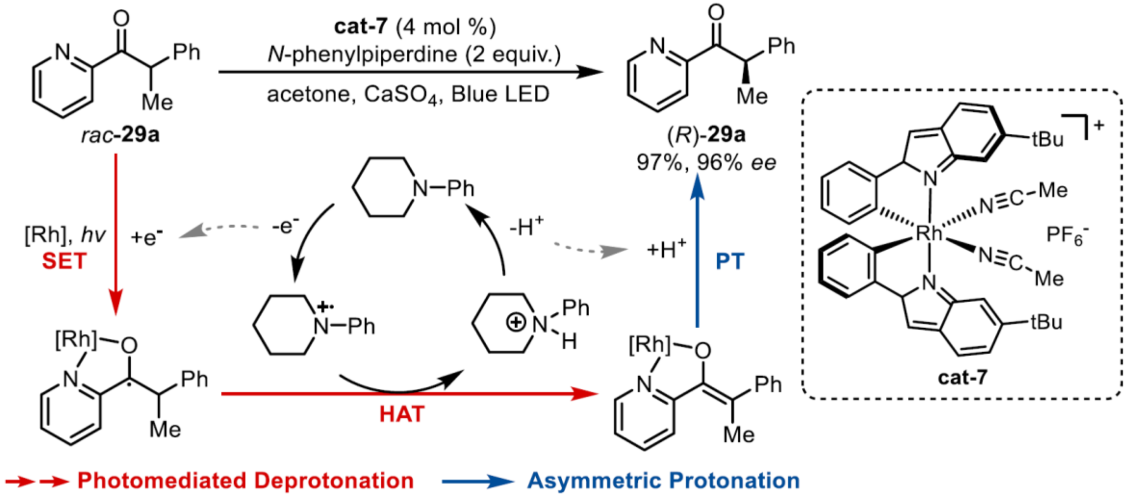
2 Linear deracemization







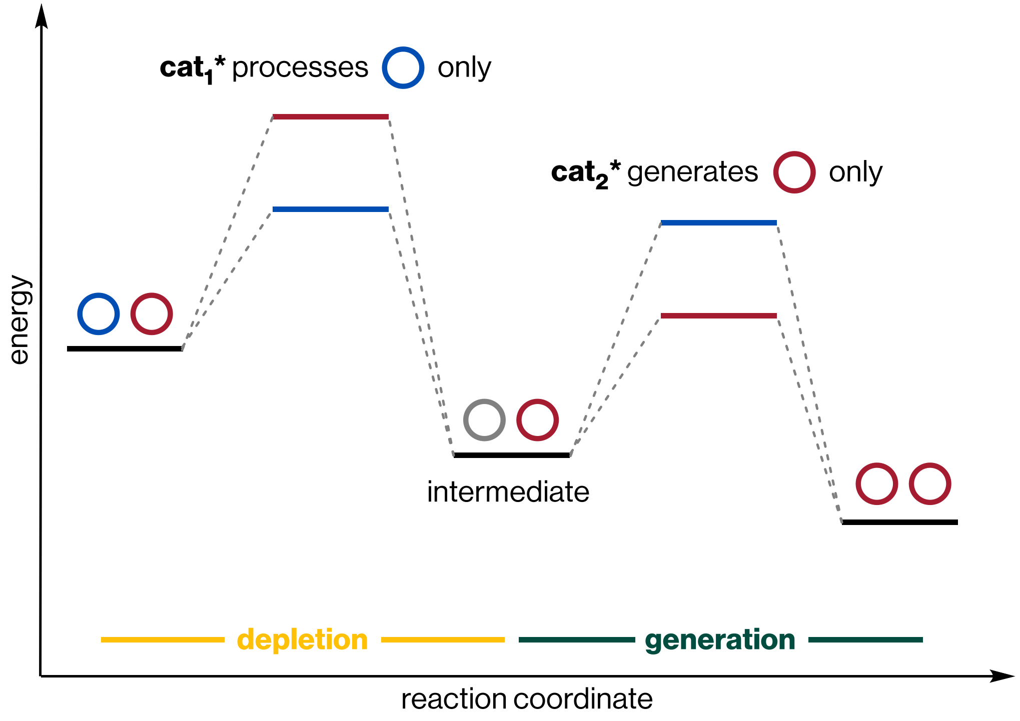
Shi & Zhou *JACS* **2015**, *137*, 10496

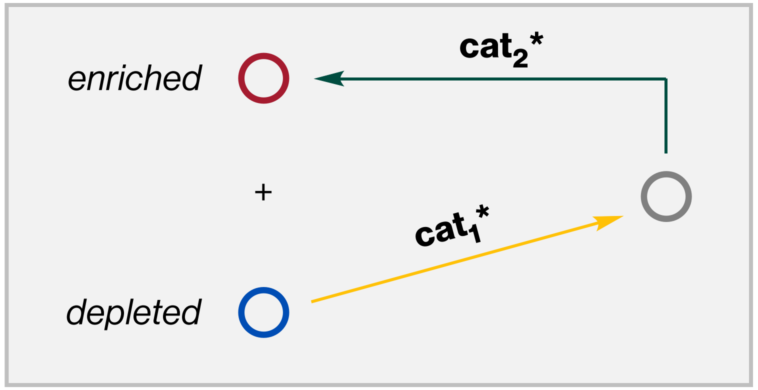


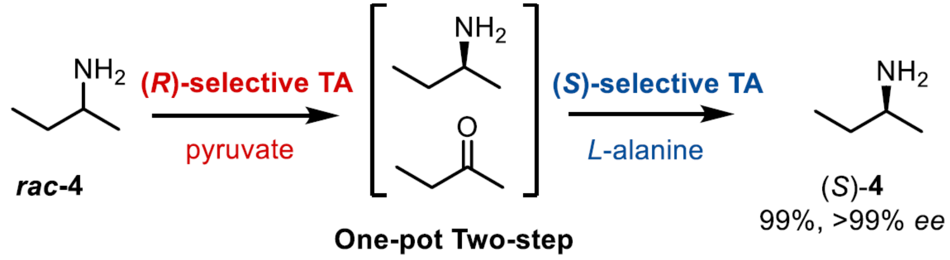
Chen & Meggers *JACS* **2021**, *143*, 13393

Stereoinvertive deracemization: selective depletion and generation

❸ Stereoinvertive deracemization

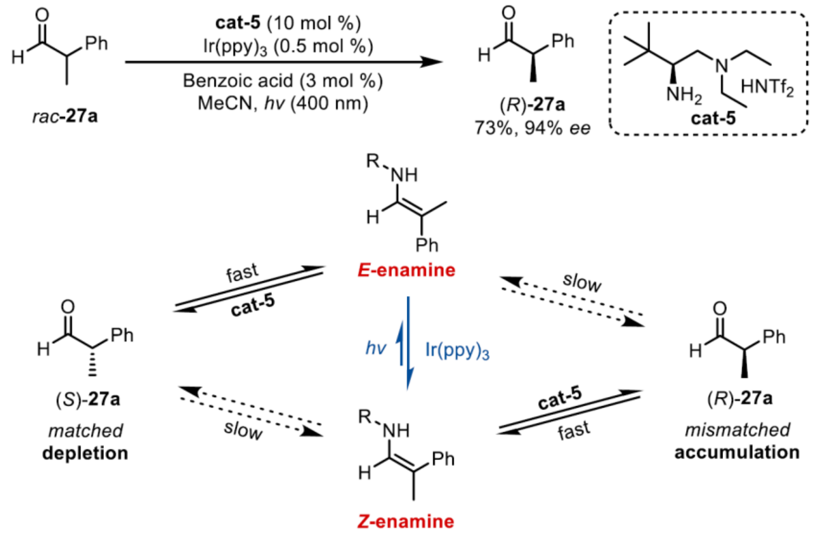






Switching sequence of enzyme addition leads to opposite outcome

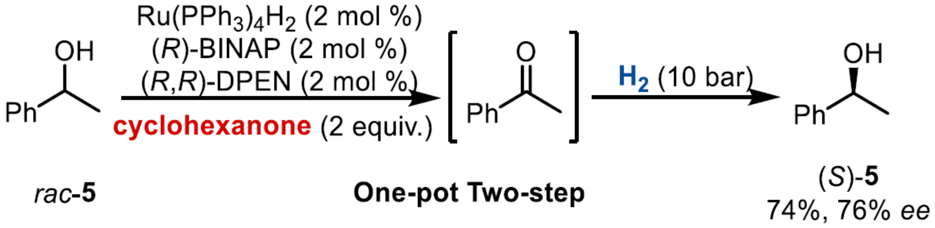
Kroutil *Eur. J. Org. Chem.* **2009**, *14*, 2289



Luo *Science* **2022**, *375*, 869

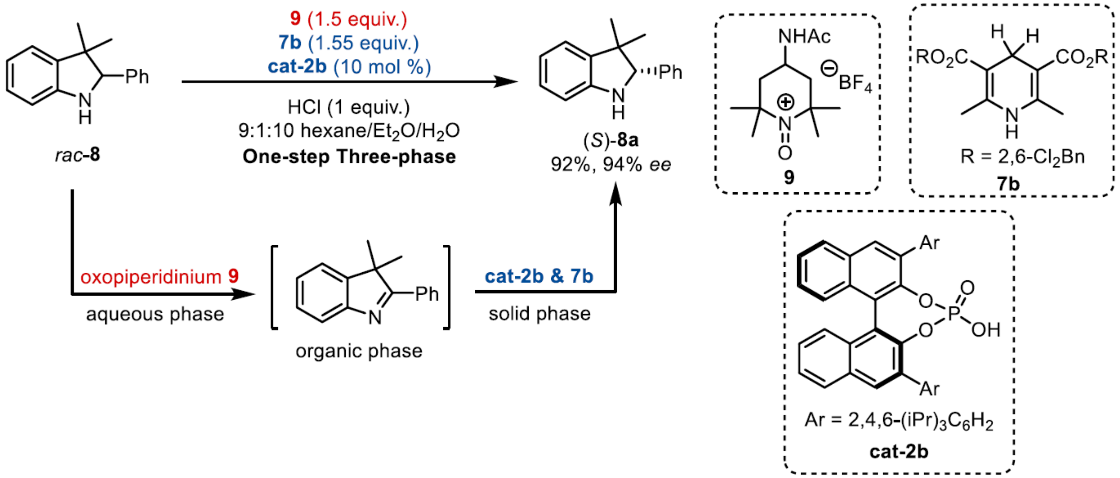
Compartmentalization of depletion/generation is key to avoid annihilation

Temporal separation



Williams *Chem. Comm.* **2005**, 5578

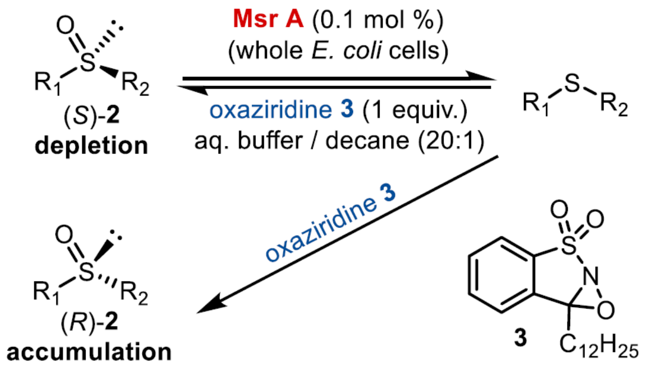
Spatial/phase separation



Toste *JACS* **2013**, *135*, 14090

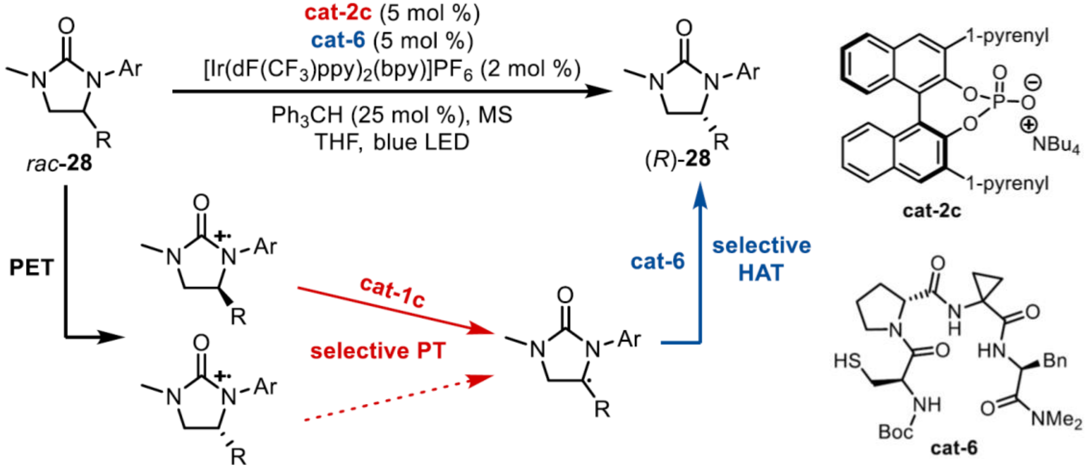
Compatible reagents

Chemo/enzymatic separation



Míšek *ACIE* **2018**, *57*, 19849

Mechanistic differentiation



Knowles & Miller *Science* **2019**, *366*, 364

Access to enantioenriched atropisomers through biocatalytic deracemization

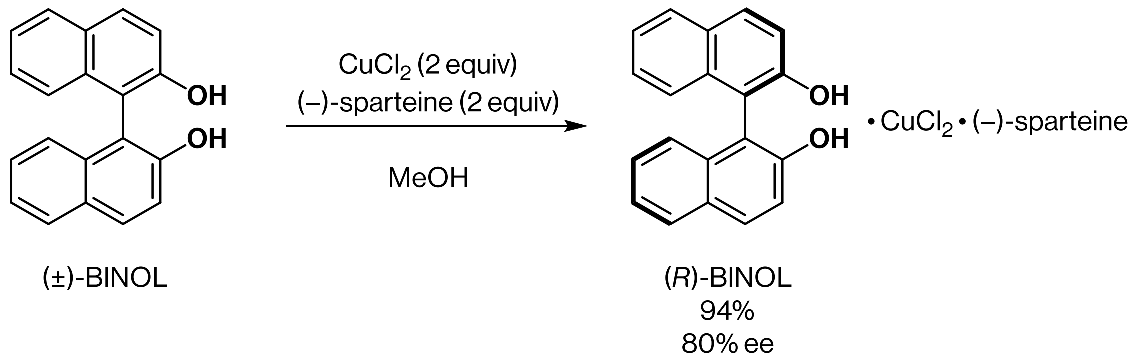
Casey B. Roos,[‡1] S. Luke Schulert, [‡1,2] Lara E. Zetzsche,[1,2] Spencer E. McMinn,[3] Angela E. Cheong,[1] Eunjae Shim,[1] Eugene E. Kwan,[3] and Alison R. H. Narayan[1,2,4*]

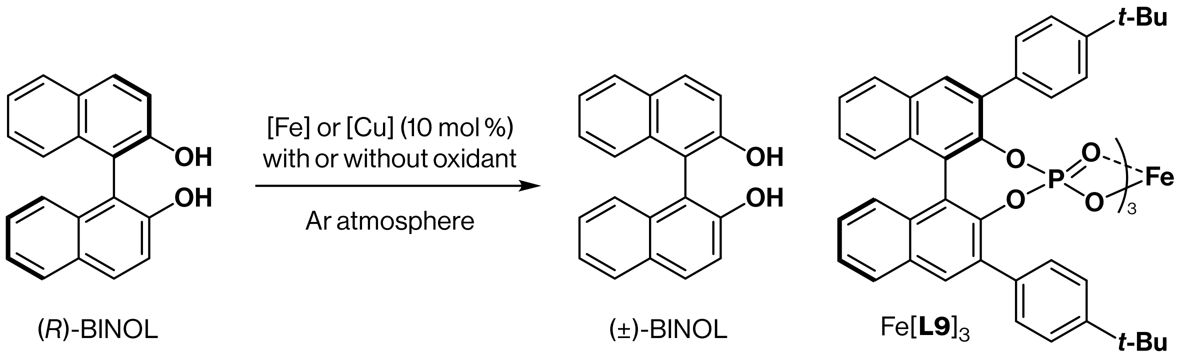
[1]Life Sciences Institute, [2]Program in Chemical Biology, University of Michigan, Ann Arbor, Michigan, United States, [3]Process Research and Development, Merck & Co. Inc., Rahway NJ, USA, [4]Department of Chemistry, University of Michigan, Ann Arbor, Michigan, United States

[‡]These authors contributed equally to this manuscript.

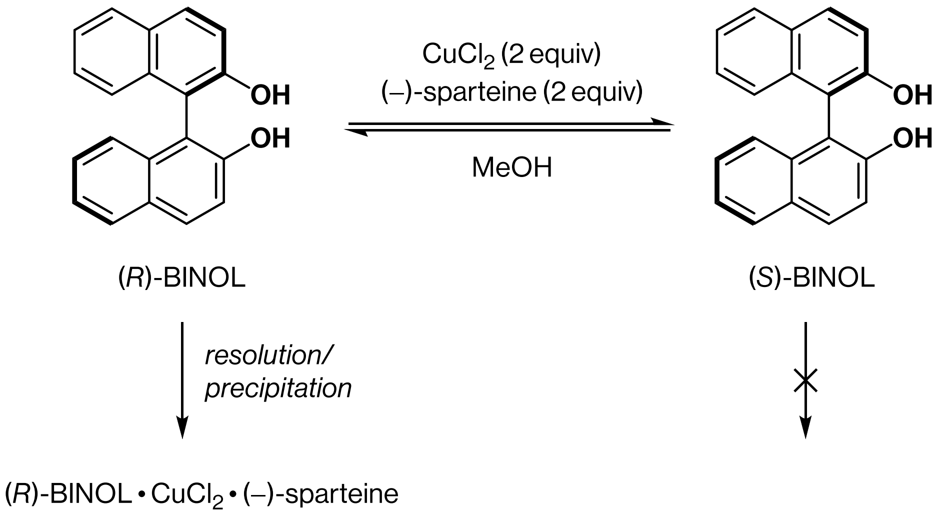
https://doi.org/10.26434/chemrxiv-2025-6s0hk

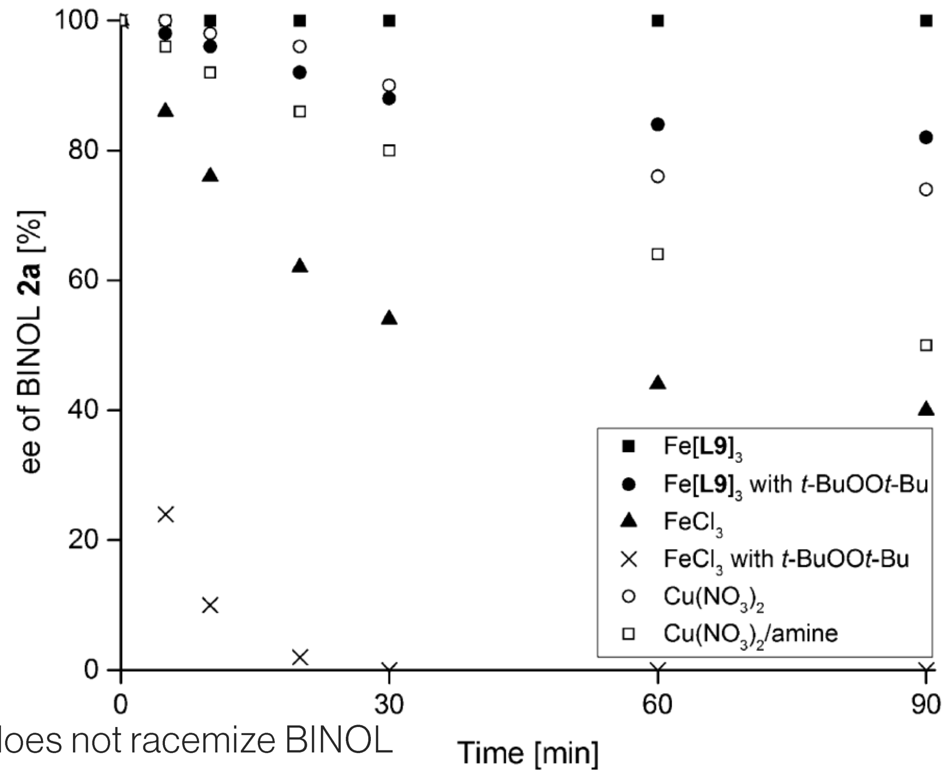
Copper salts are found to (de)racemize binaphthols





Mechanistic proposal





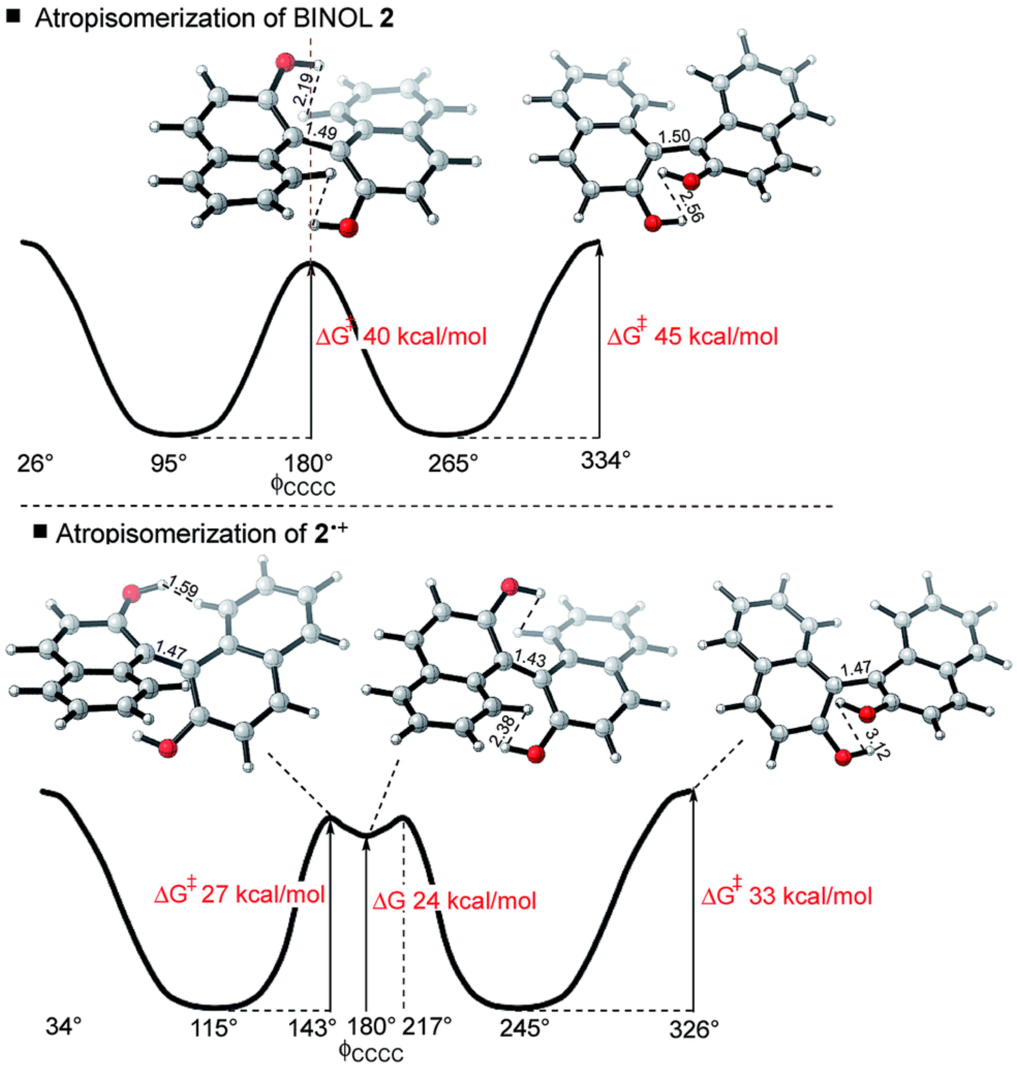
$FeCl_2$ does not racemize BINOL

Brussee *Tet. Lett.* **1983**, *24*, 3261
Kocovsky *J. Org. Chem.* **1993**, *58*, 4534

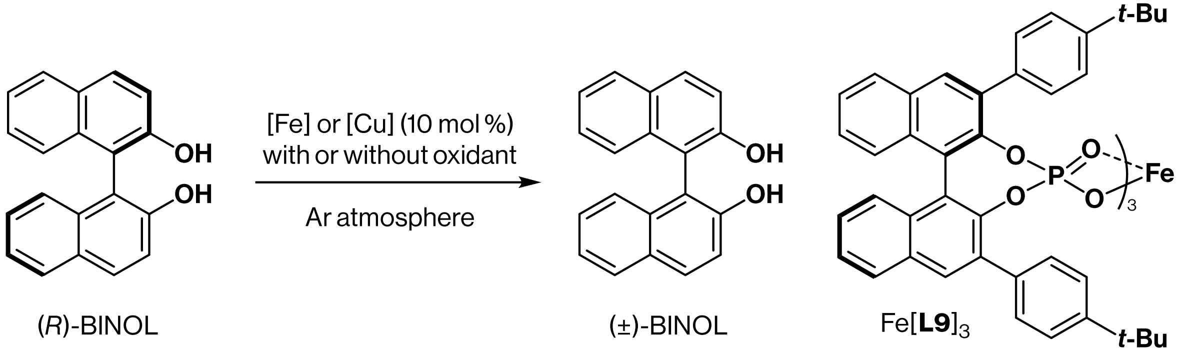
Toste & Pappo *J. Am. Chem. Soc.* **2016**, *138*, 16553

Binaphthols racemize via single-electron oxidation

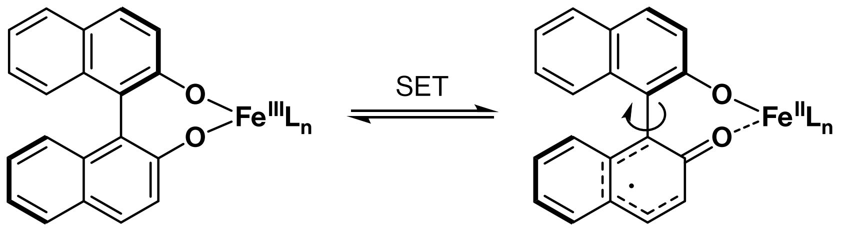
Computed barriers to rotation



Paton *Chem. Sci.* **2019**, *10*, 2285



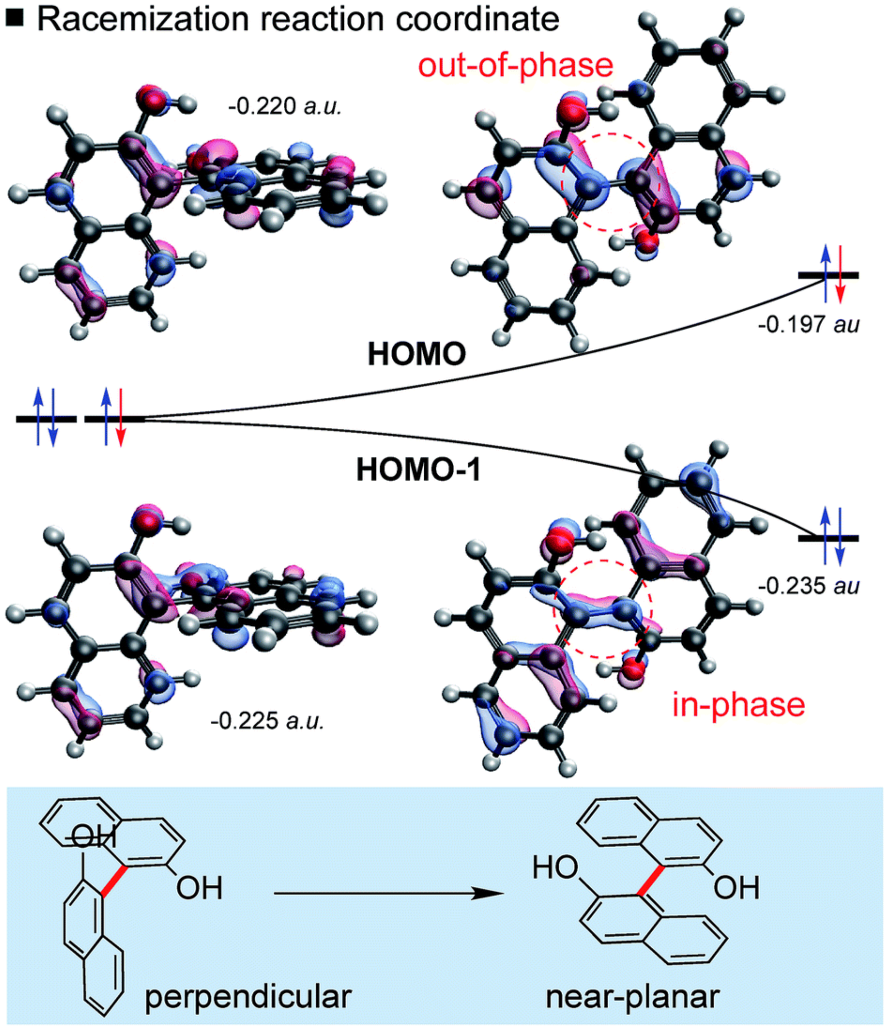
Mechanistic proposal

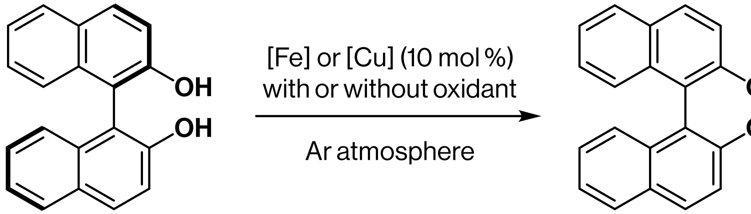


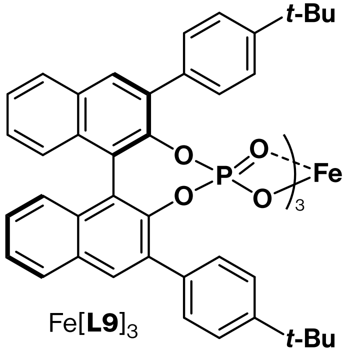
Toste & Pappo *J. Am. Chem. Soc.* **2016**, *138*, 16553

Binaphthols racemize via single-electron oxidation

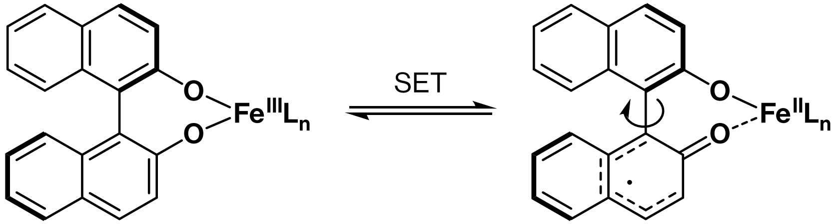
Computed barriers to rotation







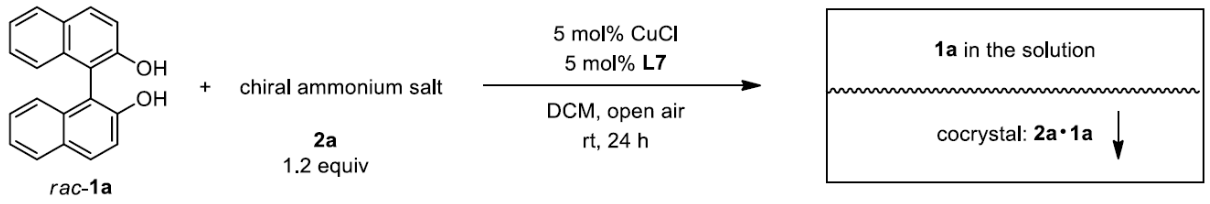
Mechanistic proposal



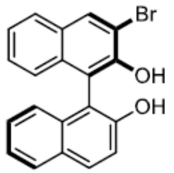
Paton *Chem. Sci.* **2019**, *10*, 2285

Toste & Pappo *J. Am. Chem. Soc.* **2016**, *138*, 16553

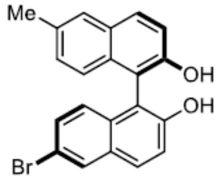
Racemization provides a strategy for catalytic dynamic thermodynamic resolution



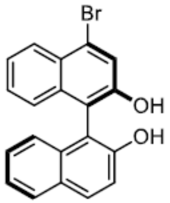
As with a classical resolution, scope is expectedly limited



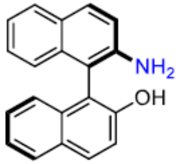
(*R*)-**1p**
67% yield, 67% *ee*



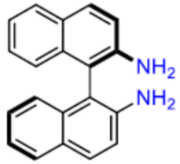
(*R*)-**1q**[b]
63% yield, 90% *ee*



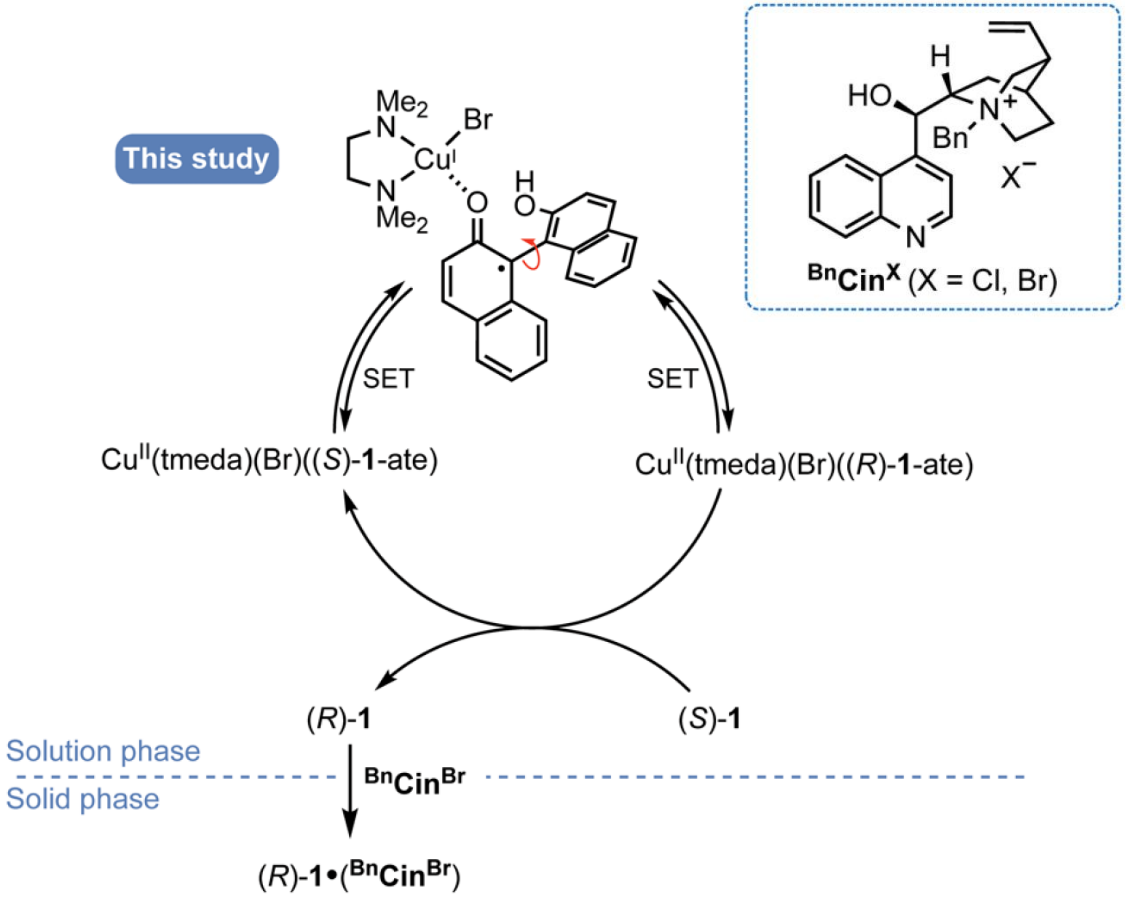
(*R*)-**1r**
N. P.



(*R*)-**1s**
71% yield, 98% *ee*



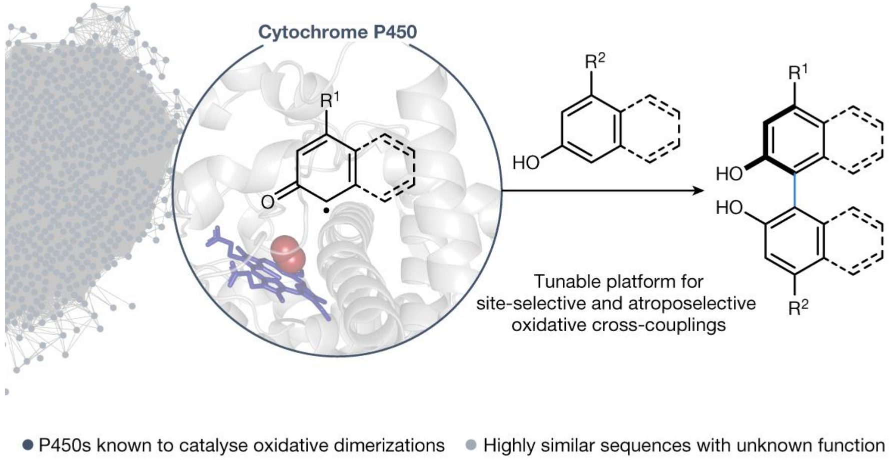
(*R*)-**1t**[c]
61% yield, >99% *ee*

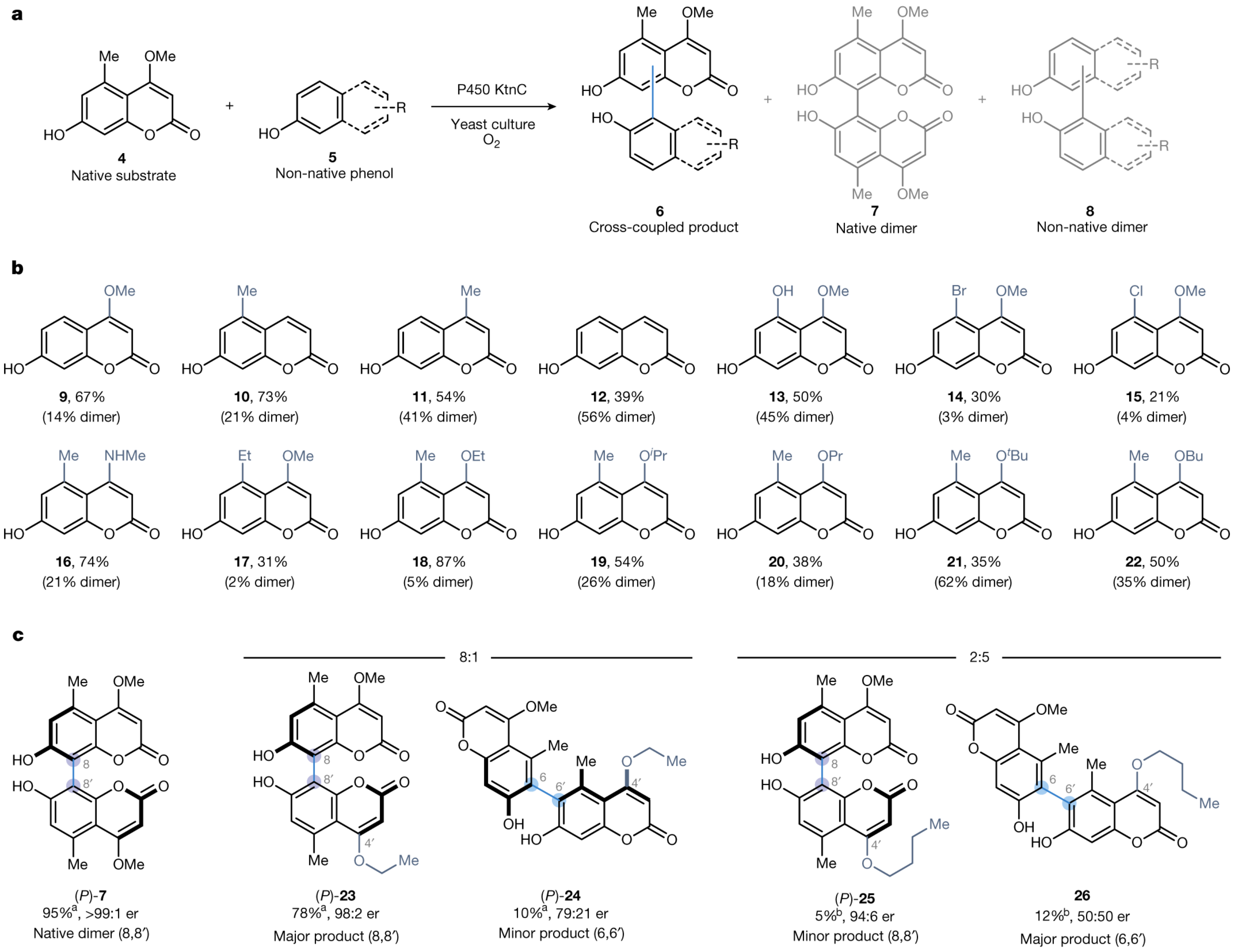


Zhu *JACS Au* **2024**, *4*, 502

Pappo & Waldvogel *Org. Lett.* **2024**, *26*, 2129

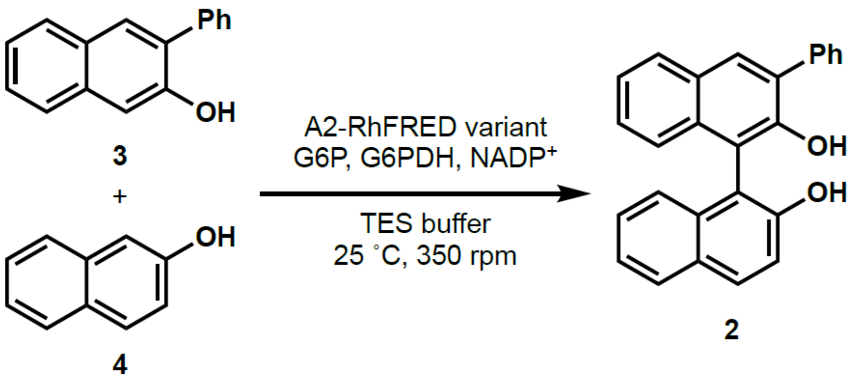
Fungal Cytochrome P450s are shown to mediate oxidative biaryl cross-couplings





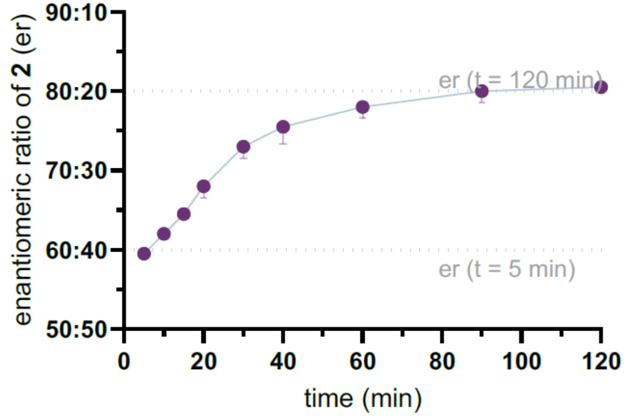
Bacterial CYP158A2 variant is identified to produce enantioenriched BINOL

Reaction scheme



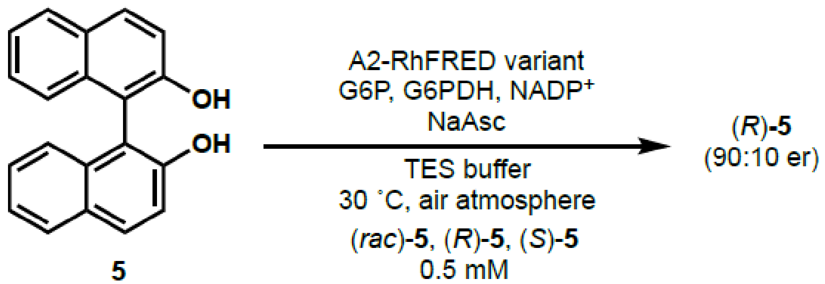
• *enantiomeric ratio of **2** increases over time* •

Enantiomeric ratio (2) vs time



Is this due to a resolution or deracemization?

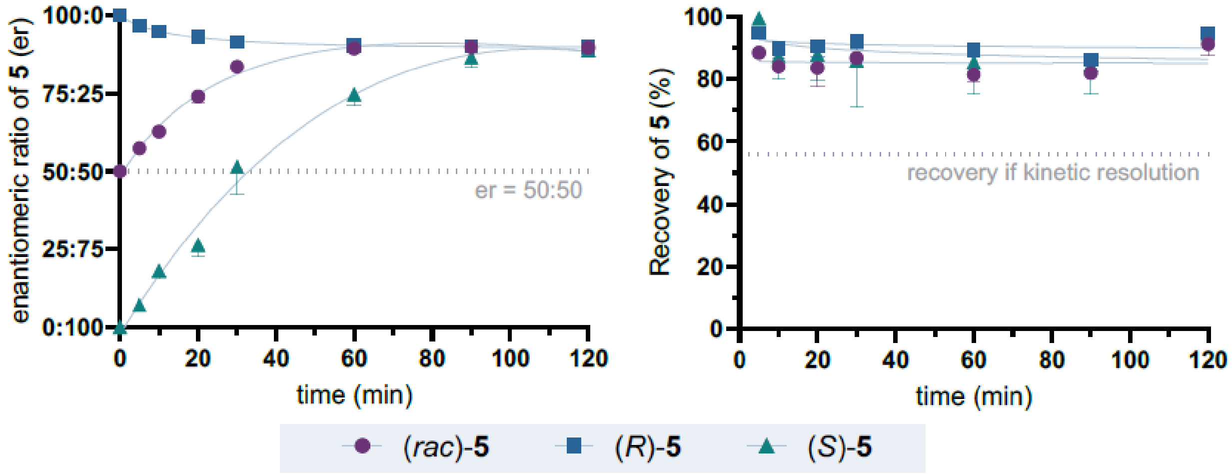
Reaction scheme



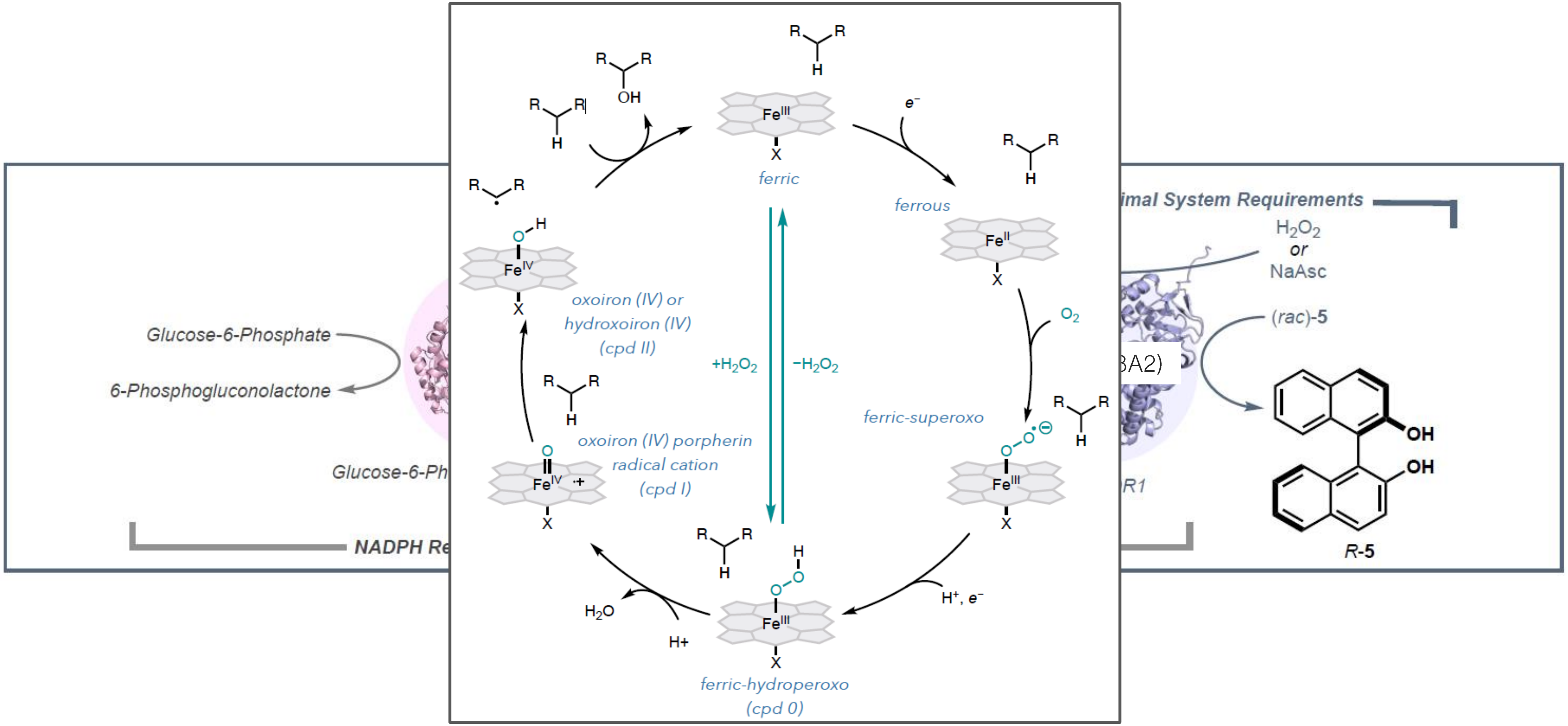
BINOL	*er (t = 0 min)*	*er (t = 2 h)*	*% recovery (t = 2 h)*
(*R*)-**5**	>99:1	90:10	95
(*S*)-**5**	<1:99	89:11	94
(*rac*)-**5**	50:50	90:10	91

Enantiomeric ratio (er) vs time — *Recovery (%) vs time*

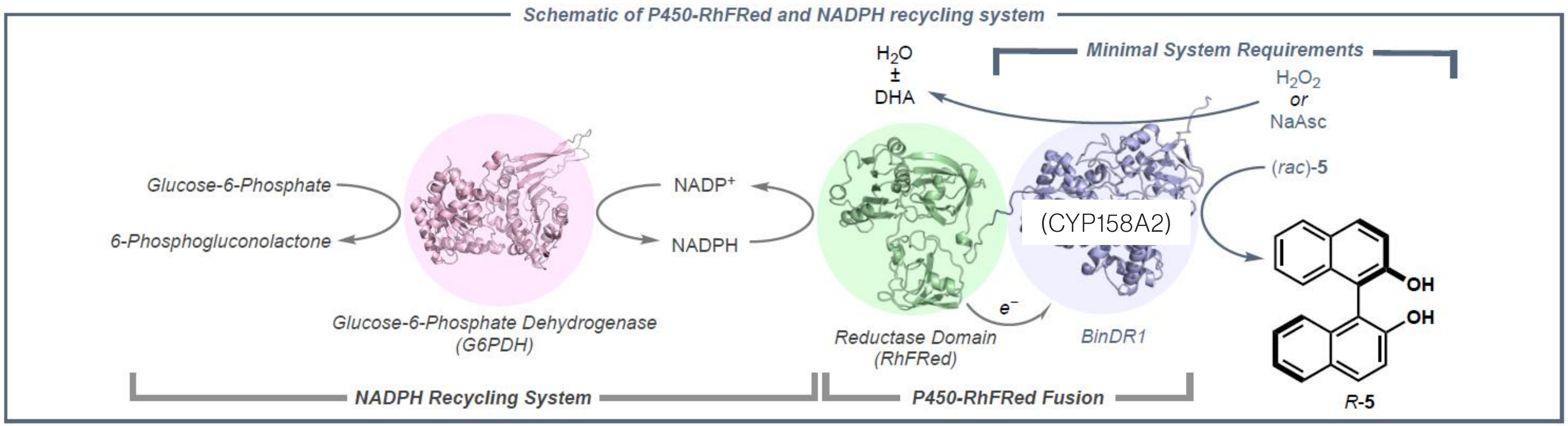
• *enantiomeric ratio of **5** converges at 90:10 er* • • *recovery of **5** remains high* •

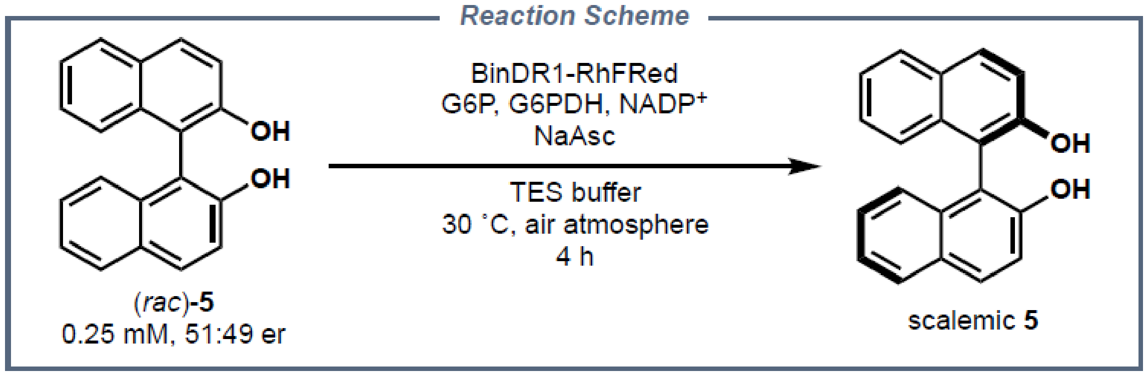


Only P450, ascorbate, and O_2 are required for deracemization



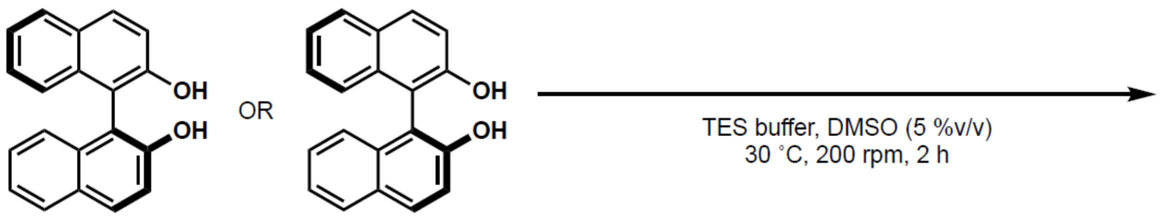
Only P450, ascorbate, and O_2 are required for deracemization





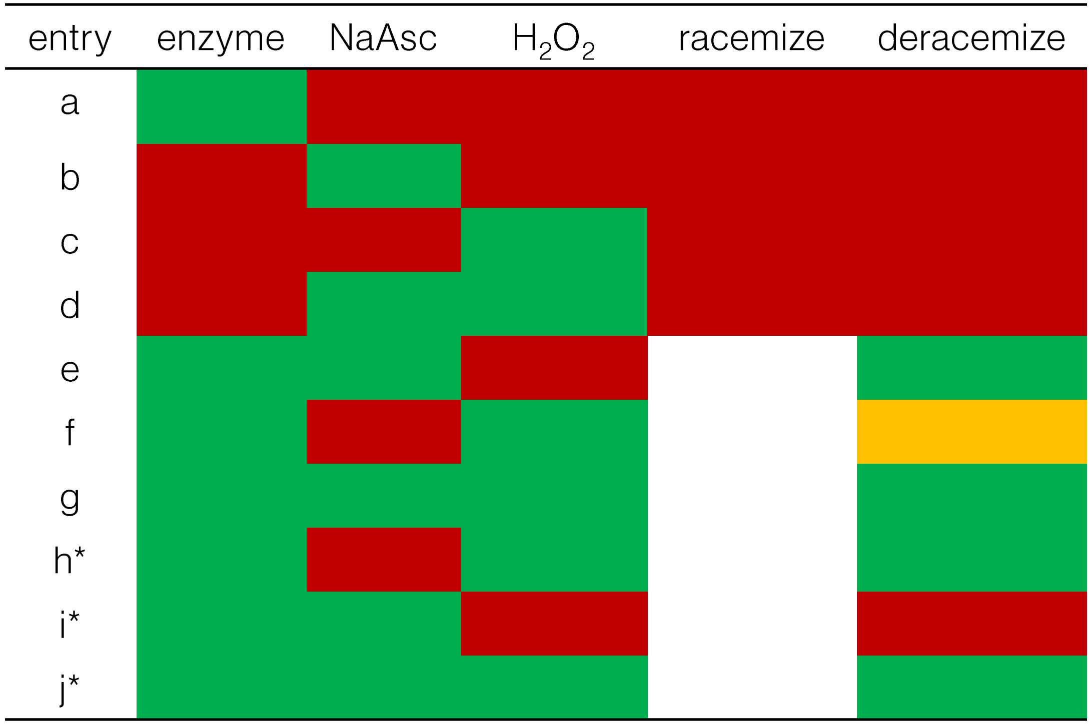
Entry	Change in conditions	Recovery (%)	er
1	no change	85	93:7
2	− BinDR1-RhFRed	92	54:46
3	+ BinDR1 − RhFRed	>95	91:9
4	+ BinDR1 − RhFRed − NADPH recycling system	87	91:9

Control experiments attempt to delineate the role of presumed oxidant and reductant



250 uM (*rac*)-**5** 50:50 er

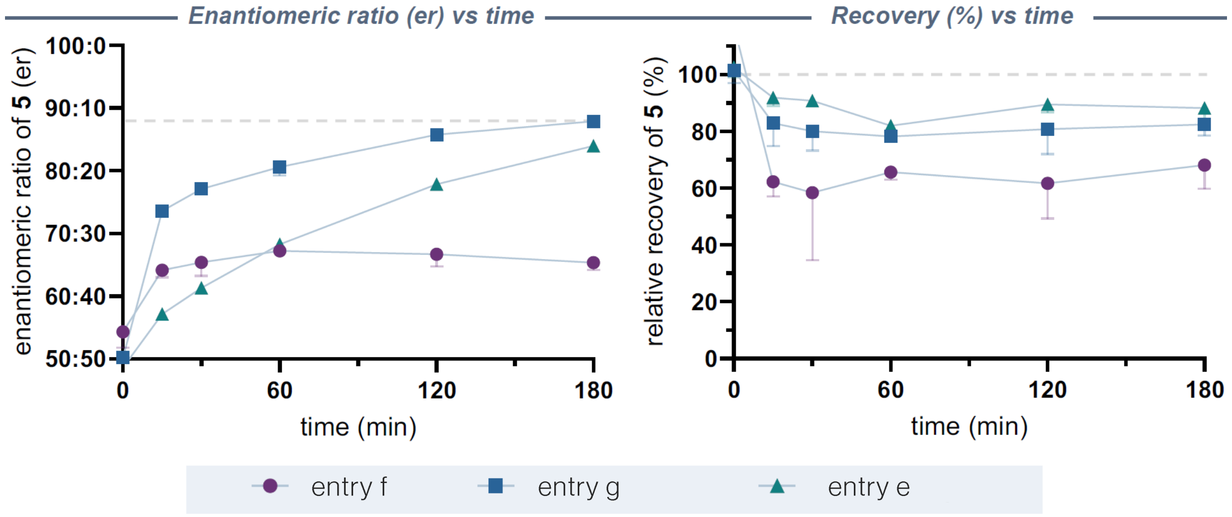
250 uM (*S*)-**5** <1:99 er



*Anaerobic conditions

Addition of a peroxide scavenger shuts down racemization

With enzyme, exogenous peroxide alone leads to loss of material, but accelerates deracemization with ascorbate

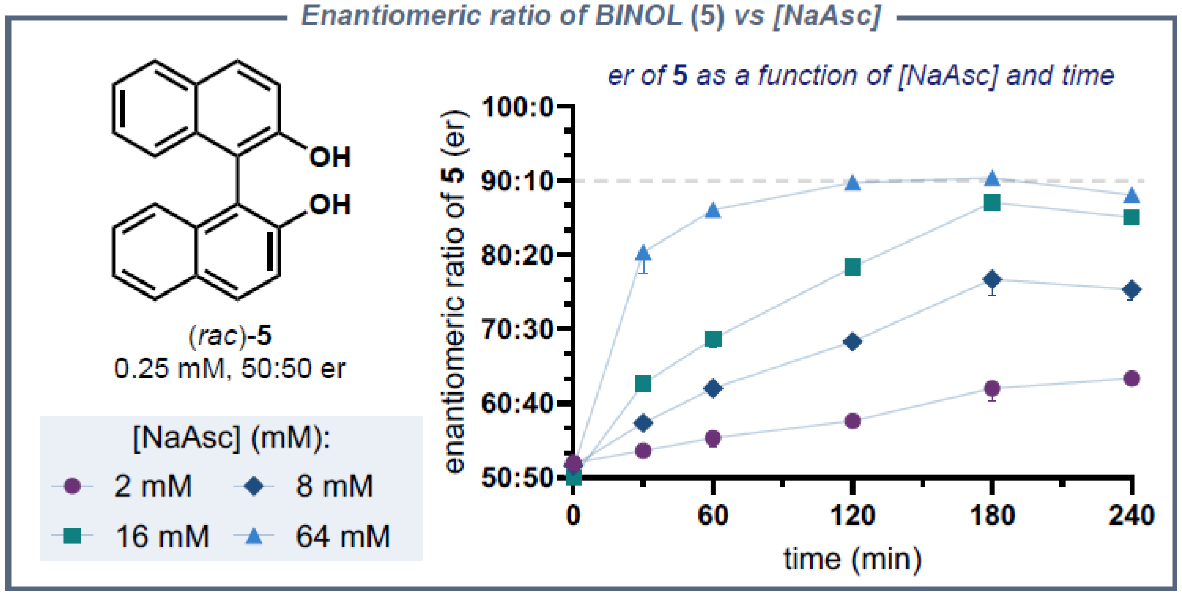


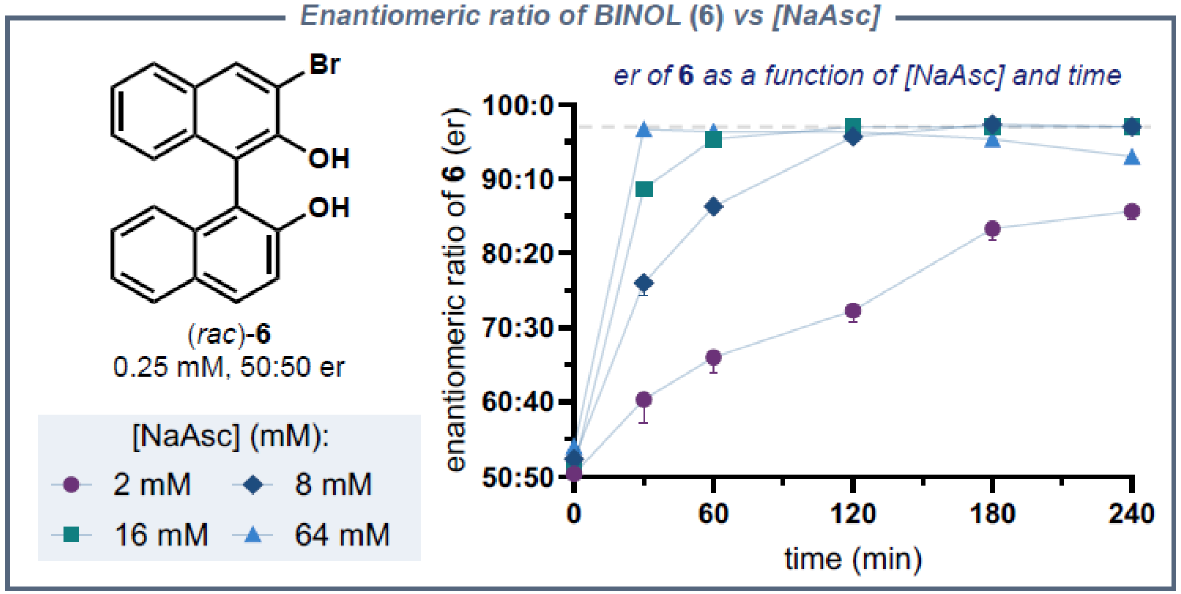
Interpretations

1. Peroxide is essential
2. Too much peroxide + oxygen leads to BINOL decomposition

Enantioenrichment depends on ascorbate concentration

Higher [NaAsc] leads to faster deracemization and higher e.r. at 240 min

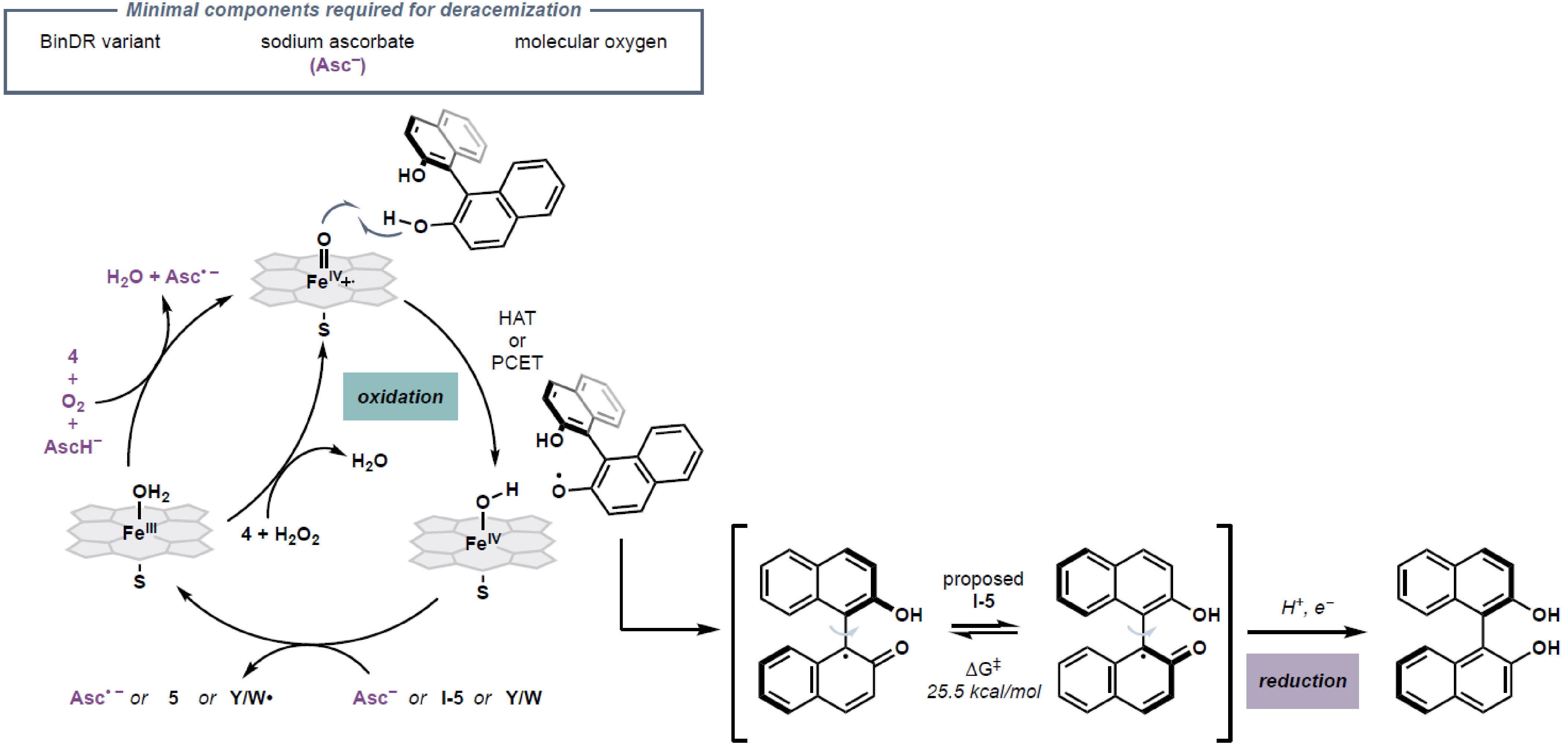




Interpretations/questions

1. “we observed that increasing the concentration of NaAsc from 2 mM to 16 mM led to both a faster enrichment and a greater final enantiomeric ratio.” (?)
2. Why is there erosion of selectivity after 180 min? Non-ascorbate-dependent background racemization/resolution?
3. “Based on a number of experiments, we do not have any evidence to support the idea that ascorbate, a chiral additive, plays a role in determining selectivity of this reaction.” (?)
4. Could ascorbate be exerting selectivity in the generation phase?

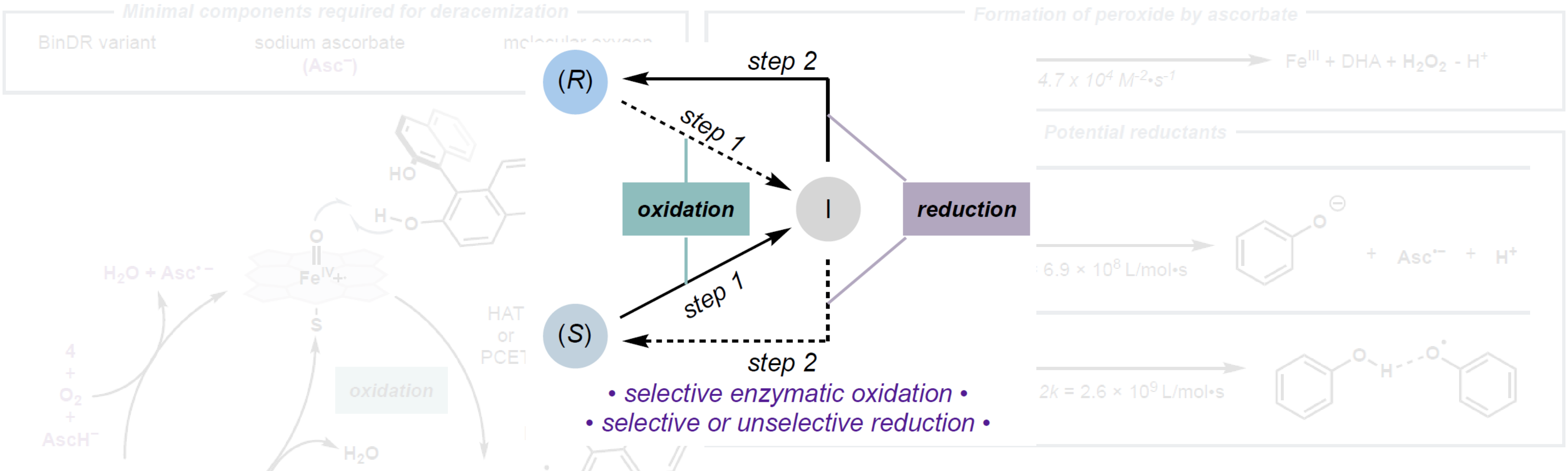
Mechanistic proposal implicates heme-dependent racemization



Enzyme is proposed to compartmentalize the oxidation step

Different enzyme variants afford different ee's, consistent with enzyme-driven depletion selectivity

Mechanistic proposal implicates heme-dependent racemization

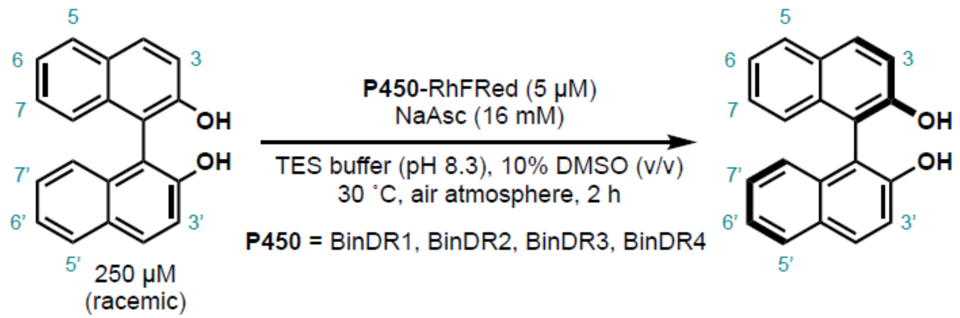


To achieve enantioenrichment through a cyclic deracemization process, one or both of the discrete mechanistic steps must exhibit stereochemistry-based selectivity. As the proposed redox reactions could take place in the inherently chiral enzyme active site, either (or both) the oxidation and reduction steps could be enantioselective, providing a plausible mechanistic rationale for deracemization. We anticipate that selectivity is derived directly from the P450, as variants can provide different final enantiomeric ratios on the same BINOL substrate. Based on a number of experiments, we do not have any evidence to support the idea that ascorbate, a chiral additive, plays a role in determining selectivity of this reaction.

Different enzyme variants afford different ee's, consistent with enzyme-driven depletion selectivity

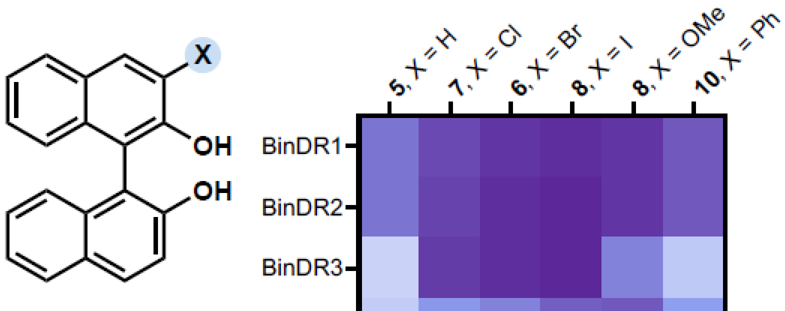
Enzyme variants display different selectivity profiles

Reaction scheme

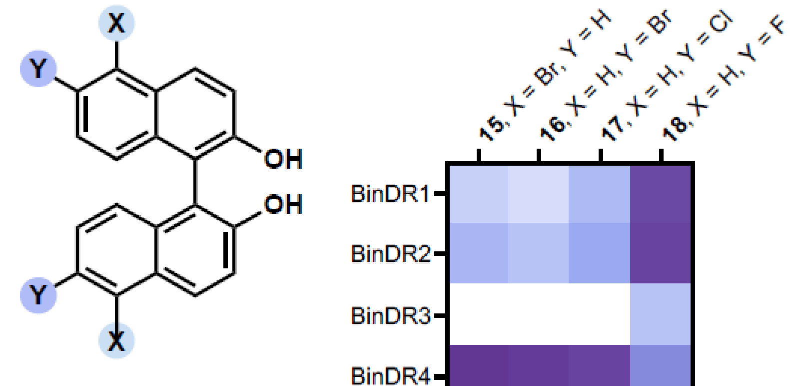


Comparison of BinDR variants

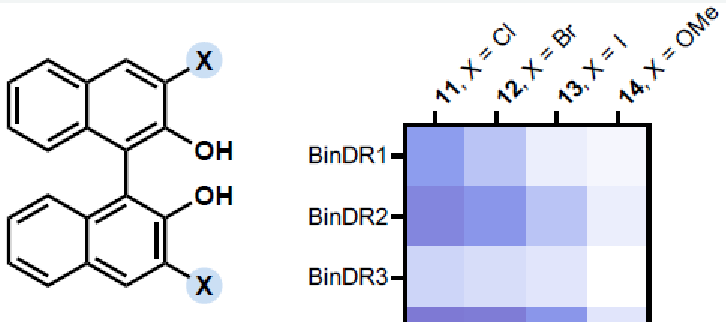
Variant/ Mutations	A83L	S181G	A290M	P45E	R90Q	A245Q	R81S	M75F	H183K	I241L
BinDR1	●	●	●	●	●	●				
BinDR2	●	●	●	●	●	●	●			
BinDR3	●	●	●	●	●					
BinDR4	●	●	●			●		●	●	●



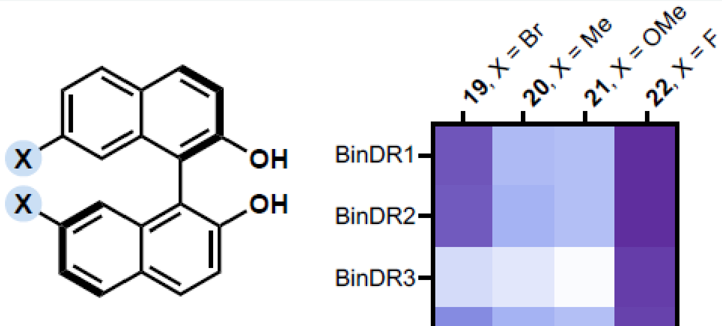
3-substitution



5,5'- and 6,6'-substitution



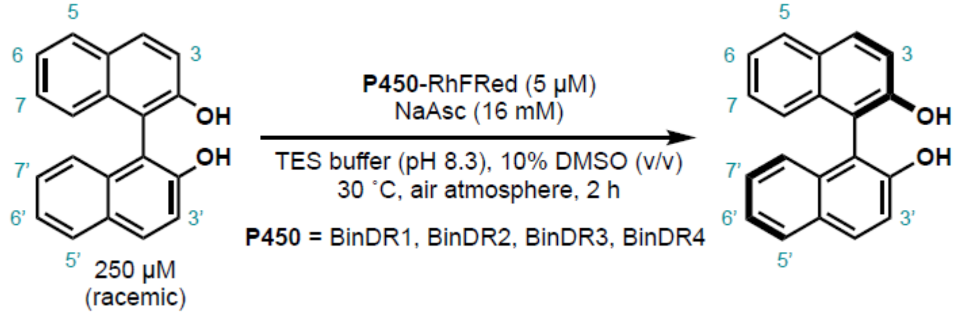
3,3'-substitution



7,7"-substitution

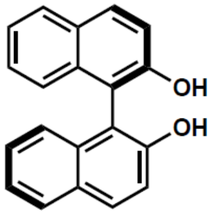
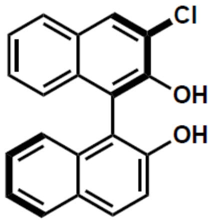
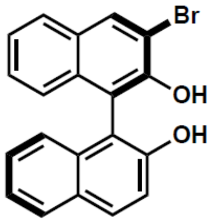
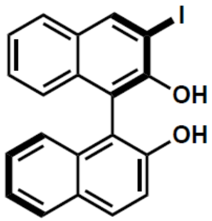
Enzyme variants display different selectivity profiles

Reaction scheme

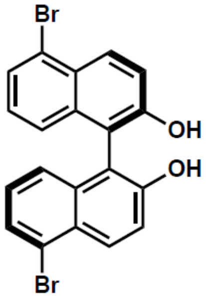
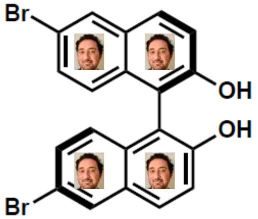
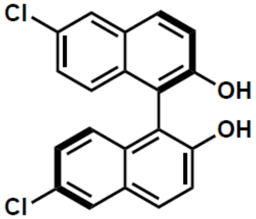
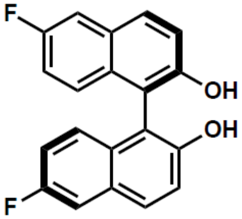


Comparison of BinDR variants

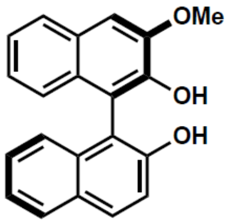
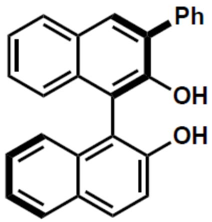
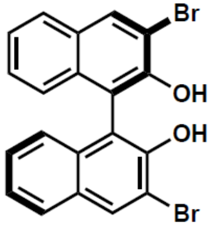
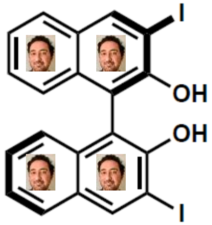
Variant/ Mutations	A83L	S181G	A290M	P45E	R90Q	A245Q	R81S	M75F	H183K	I241L
BinDR1	●	●	●	●	●	●				
BinDR2	●	●	●	●	●	●	●			
BinDR3	●	●	●	●	●					
BinDR4	●	●	●			●		●	●	●

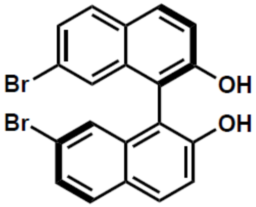
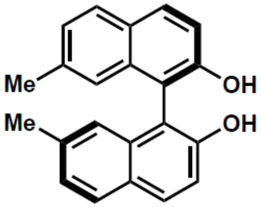
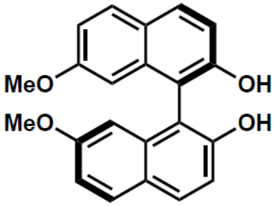
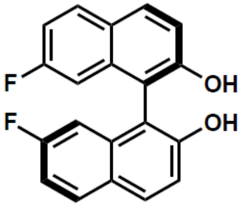
variant	(R)-5[a]	7	(R)-6[a]	(R)-8[a]
BinDR1	84% yield, 88:12 er	>95% yield, 94:6 er	>95% yield, 97:3 er	>95% yield, 98:2 er
BinDR4	77% yield, 67:33 er	>95 % yield, 83:17 er	>95 % yield, 86:14 er	92% yield, 91:9 er

variant	15	(R)-16[a]	17	18
BinDR1	>95 % yield, 61:39 er	92% yield, 58:42 er	74% yield, 66:34 er	81% yield, 91:9 er
BinDR4	86% yield, 95:5 er	>95% yield, 94:6 er	87% yield, 92:8 er	80% yield, 77:23 er

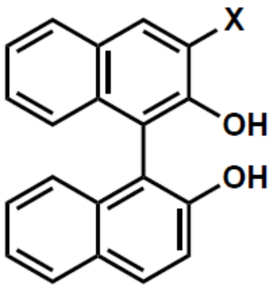
variant	9	(R)-10[a,b]	(R)-12[a]	(R)-13[a]
BinDR1	83% yield, 97:3 er	77% yield, 8:92 er	>95% yield, 43:57 er	91% yield, 48:52 er
BinDR4	91% yield, 92:8 er	75% yield, 16:82 er	>95% yield, 28:72 er	76% yield, 36:64 er

variant	19	20	21	22
BinDR1	84% yield, 89:11 er	70% yield, 67:33 er	86% yield, 66:34 er	94% yield, 97:3 er
BinDR4	89% yield, 78:22 er	75% yield, 69:31 er	85% yield, 66:34 er	87% yield, 93:7 er

Wildtype CYP158A2 has low but nonzero deracemization activity, unlike other P450s

Reaction scheme



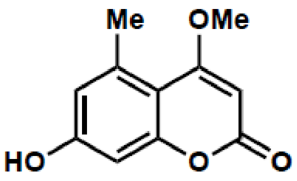
X = H
(*rac*)-**5**, (*R*)-**5**, (*S*)-**5**

X = Br
(*rac*)-**6**, (*R*)-**6**, (*S*)-**6**

A2-RhFRed wild-type
G6P, G6PDH, $NADP^+$
NaAsc
→
TES buffer
30 °C, air atmosphere

scalemic
5 or **6**

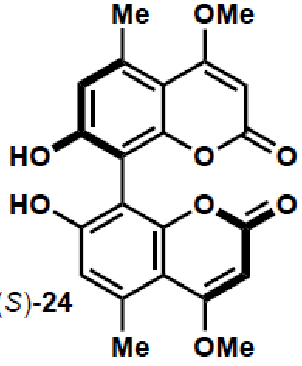
BINOL **5**	er (*t* = 2 *h*)	BINOL **6**	er (*t* = 2 *h*)
(*R*)-**5**	93:7	(*R*)-**6**	85:15
(*S*)-**5**	7:93	(*S*)-**6**	30:70
(*rac*)-**5**	53:47	(*rac*)-**6**	59:41



23

- *or* - (*rac*)-**24**

wild-type KtnC
→
His⁻ 6% galactose media
30 °C, 300 rpm, 2 d



(*S*)-**24**

Entry	*Time (h)*	*er 24 Dimerization*	*er 24 Deracemization*
1	24	>99:1	52:48
2	48	>99:1	52:48

Wishlist and unresolved questions

1. Could the authors have provided a computational model for selectivity in the enzyme-catalyzed step?
2. Is it possible to characterize the reduction (generation) step more rigorously?
3. Longer monitoring times for NaAsc-dependent reaction profiles could illuminate contributions from other processes
4. Can the oxidized species be trapped? If excess peroxide is harmful, could they characterize overoxidation products?
5. Is there a way to study the heme-dependent HAT and reverse HAT steps in isolation?
6. Is it clear why the enzyme is beating microscopic reversibility?

(Why) Is this an interesting/notable advance?

1. Extends catalytic deracemization to atropisomers
2. Integrates enzyme-catalyzed HAT into a deracemization scheme
3. HAT chemistry applied to indirectly access conformational isomerization (is this extensible past BINOL?)
4. What are the implications for small-molecule catalysis?